Advanced Monograph Series

Mathematical Theory Of Radon Transforms

A rigorous study of the Radon transforms

and its technical foundations

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This monograph is intended as a rigorous analytical exposition of the theory of Radon transforms, including its mathematical foundations, analytical framework, and associated applications in integral geometry, partial differential equations, and inverse problems.

The author does not claim original discovery of the mathematical results presented herein. This work is a systematic and rigorous exposition of established theory, compiled for clarity, accessibility, and deeper analytical understanding. All primary references and source materials are listed in the bibliography section at the end of this work.

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Preface

This book exists for a simple reason. I wanted a solid and rigorous book on Radon transforms that builds everything properly from the ground up.

Most of the material available online is too scattered, too brief, or assumes too much without connecting ideas clearly. Radon transforms are too beautiful and important to be treated like disconnected notes across different sources. I wanted one place where the theory stands together in a complete and meaningful way.

The approach of this book is simple: prove everything. Nothing important is taken for granted. Every theorem is placed carefully, every result is justified, and examples are included so the reader can work with the concepts instead of only reading them.

This book is mainly for serious readers, graduate students, and undergraduates stubborn enough to willingly enter this territory. Since the subject touches several areas of mathematics, a strong undergraduate foundation in calculus, linear algebra, analysis, PDEs, and related topics will help greatly. A preliminaries section and appendix are also included, and my honest suggestion is simple: read the book in order.

I do not claim original discovery of the results presented here. This work is an effort to compile and present the existing theory in a rigorous and readable form, with proper references given at the end.

If this book helps even one serious reader understand Radon transforms more deeply, then it has done its job.

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Preliminaries

Before we begin our discussion of Radon transforms, it becomes essential to know certain properties. Starting off with Linear algebra, for any two given vectors A, B such that their dimensionality is given as $dim(A) = dim(B) = n \times 1$ and $A = [a_i], B = [b_i]\ \forall\ 1 \leq i \leq n$, dot product of the two is given as:

$$A \cdot B = A^T B = \sum_{i=1}^{n} a_i b_i$$

Now, considering $A, B \in M_{n \times n}(\mathbb{R})$, the following properties hold:

(a) $(AB)^T = B^T A^T$
(b) $(AB)^2 = ABAB \neq A^2 B^2$ (Generally, matrix multiplication is non-commutative.)
(c) $det(AB) = det(A)\, det(B)$
(d) $det(cA) = c^n\, det(A)$ (Where, c is some positive real number)
(e) $A^{-1} = \frac{Adj(A)}{det(A)}$. Thus, if $det(A) = 0$ matrix A doesn't have an inverse meaning A is a singular matrix.
(f) If $A^{-1} = A^T$, then $det(A^{-1}) = det(A)$.

Other than this, knowing Matrix integrals is also quite important. Let, $X = (x, y)$, then the following integrals are equivalent by nature.

$$\int_{X \in \mathbb{R}^2} f(X)\, dX = \int_{x \in \mathbb{R}} \int_{y \in \mathbb{R}} f(x, y)\, dx\, dy$$

Matrix based substitutions such as, $X = AY$ assuming $X = (x, y), Y = (m, n)$ and $A = \begin{bmatrix} cos\, t & -sin\, t \\ sin\, t & cos\, t \end{bmatrix}$ will result in $dX = det(A)\, dY = (cos^2 t + sin^2 t)\, dy = dY$, ultimately leading to the following result:

$$\int_{X \in \mathbb{R}^2} f(X)\, dX = \int_{Y \in \mathbb{R}^2} f(AY)\, dY$$

$$\int_{X\in\mathbb{R}^2} f(X)\,dX = \int_{m\in\mathbb{R}}\int_{n\in\mathbb{R}} f(m\cos t - n\sin t\,, m\sin t + n\cos t)\,dn\,dm$$

So, in general, for every integral substitution $U = AV, dU = det(A)\,dV$, such that $det(A) > 0$, $U = [u_i], V = [v_i]\ \forall i \le n$ will result in the following integral identity:

$$\int_{U\in\mathbb{R}^n} f(U)\,dU = det(A)\int_{V\in\mathbb{R}^n} f(AV)\,dV$$

Similarly, knowing about partial derivatives will also be quite important. We will considering the usual function $f(x,y)$, partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are defined as follows:

(A) $\frac{\partial f}{\partial x} = \frac{f(x+h,y)-f(x,y)}{h}$
(B) $\frac{\partial f}{\partial y} = \frac{f(x,y+h)-f(x,y)}{h}$

Considering $h \to 0$ in both the cases. Also, it would be beneficial to know the Euler's trick too:

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial^2 f}{\partial y\partial x}$$

The above holds considering f is a differentiable function in $\mathbb{R}^2$ domain.

Other than this, knowing the Fubini's trick on double integrals will also prove to be quite helpful given f is an integrable function in $x \in A, y \in B$ domain. (Let, A, B be some domains on the real line):

$$\int_{y\in B}\int_{x\in A} f(x,y)\,dx\,dy = \int_{x\in A}\int_{y\in B} f(x,y)\,dy\,dx$$

Similarly, knowing the delta-δ function will be crucial in dealing with Radon transforms, and its definition is given as follows:

$$\delta(x) = \begin{cases} \infty\,; x = 0 \\ 0\,; x \in \mathbb{R} \,/\, \{0\} \end{cases}$$

It's integral identities are as follows (Assume, f to be some well-defined function):

(A) $\int_{\mathbb{R}} \delta(x)\, dx = 1$
(B) $\int_{\mathbb{R}} f(x)\delta(x)\, dx = f(0)$
(C) $\int_{\mathbb{R}} f(x)\delta(x-a)\, dx = f(a)$

Other than this, knowing a bit of Graph theory would also be helpful in the later sections. Let $G(V, E)$ be some graph G with set of vertices V and set of edges E, then the total number of connections obtained by joining all vertices is given as follows, considering $|V| = m$:

$$max(|E|) = \binom{m}{2} = \frac{m!}{(m-2)!\,2!} = \frac{m(m-1)}{2}$$

Similarly, assuming $|V| = m$ and we wish to construct a n-gon ($m > n$, $n \geq 3$). Here, E described nicely using the following set notation:

$$E = \{(v_1, v_2), (v_2, v_3), \dots, (v_n, v_1)\}$$

Assuming $V = \{v_i | 1 \leq i \leq m\}$. Here, $|E| = n$ and this is also called as C_n (Cyclic group of order n) and is directly correlated to the group $\mathbb{Z}_n$.

Similarly, if we want to construct a n-gon with all diagonals connected, then E can be described as follows:

$$E = \left\{(v_i, v_j) \middle| \forall 1 \leq i \leq n-1\,, \forall i < j \leq n\right\}$$

This conditional counting ensures there is no overlapping and all elements are traversed properly. Here, this graph can be represented as K_n (Complete graph of order n) and $|E| = \frac{n(n-1)}{2}$.

To close out the preliminaries section let us finally discuss Adjacency matrix or point-incidence matrix:

$$A(i,j) = \begin{cases} 1;\, if\ (i,j) \in E \\ 0;\, if\ (i,j) \notin E \end{cases}$$

Such that, $dim(A) = |E| \times |E| = n \times n$. This concludes the preliminaries; we can now formally discuss the theory of Radon transforms.

Notations

One of the most common notations used throughout the book is based around matrix representation. For example, $A = [a_i] \ \forall i \leq n$ simply means the following:

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix}$$

Meaning matrix A has dimensionality of $n \times 1$ (i.e. $dim(A) = n \times 1$). This is just a short-hand notation to represent an n-dimensional vector introduced in order to save time.

Another notation used quite regularly throughout this book being, an n-dimensional vector X having its cardinality written out as $|X| = n$. Do note that it simply means $dim(X) = n \times 1$ and to avoid unnecessary longevity in sentences, the notation of $|X| = n$ is adopted from hereon. Just to make it clear, the vector X can be represented as follows (here it is assumed $n > 2$ for representational purposes):

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$$

A few parameters assumed here are slightly different from the older texts and manuscripts. For instance, the principal value in Radon transforms is denoted in this textbook with P, corresponding unit matrix is denoted using U and in two-dimensional cases t is considered to represent the angle and hence $U = (cos\, t\, , sin\, t)$ will be a common assumed notation. Other than this standard R, F, M, L, H are used to denote Radon, Fourier, Mellin, Laplace and Hilbert transforms respectively. Effort has been taken to ensure situations such as $F(F)$ or $H(H)$ don't arise anywhere in all of the upcoming sections.

Introduction to Radon transforms

Let us define a two-dimensional function $f(x, y)$ such that $x, y \in \mathbb{R}$. We wish to take an integral of this function f over a given line $L : ux + vy = P$, such that the weights (u, v) contribute to a unit vector U. Here, the integration assumes all such x, y which satisfy the constraint set by L and thus, we state $x, y \in S^2$. Assuming, R to be the Radon transform and m to be the specialized distribution function that maps all x, y in the given region S accurately. All of this can be symbolically stated as follows:

$$R\{f(x, y)\}(P, u, v) = \int_L f(x, y)\, dm(x, y)$$

$$R\{f(x, y)\}(P, u, v) = \int_{x,y \in S^2} f(x, y)\, dm(x, y)$$

Introducing Dirac-delta function δ becomes pivotal in Radon transform's representation. Notice that the delta function $\delta(P - ux - vy)$ will ensure that all participating $x, y \in S$ as they directly satisfy the constraint set by L. Due to which, the Radon transform can be restated using the delta function:

$$R\{f(x, y)\}(P, u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\delta(P - ux - vy)\, dx\, dy$$

Now, the above can be rewritten in linear algebraic fashion by assuming $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $U = \begin{bmatrix} u \\ v \end{bmatrix}$. Recall that, $U \cdot X = U^T X = ux + vy$.

$$R\{f(X)\}(P, U) = \int_{X \in \mathbb{R}^2} f(X)\, d(P - U^T X)\, dX$$

Here, we introduce $A = \begin{bmatrix} u & -v \\ v & u \end{bmatrix}$. It would be good to know that, $R\{f(X)\} = R\{f(AX)\}$ and this holds true as $|det(A)| = 1$. Using A, we perform a substitution by assuming $X = AY$ and as a consequence we will have $dY = dX$. Let us define another object as $B = A^{-1}$. This means, $B = \frac{Adj(A)}{det(A)} = \begin{bmatrix} u & v \\ -v & u \end{bmatrix} = A^T$. By using all of this, we have:

$$R\{f(X)\}(P,U) = \int_{Y\in\mathbb{R}^2} f(AY)\delta(P - U^T AY)\, dY$$

$$R\{f(Y)\}(P,U) = \int_{Y\in\mathbb{R}^2} f(AY)\delta(P - (BU)^T Y)\, dY$$

Assuming, $Y = \begin{bmatrix} h \\ k \end{bmatrix}$, we simplify the above expression. Observe that, $BU = \begin{bmatrix} u & v \\ -v & u \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u^2 + v^2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $(BU)^T Y = h$.

$$R\{f(x,y)\}(P) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(uh - vk, vh + uk)\delta(P - h)\, dk\, dh$$

But by setting $(u, v) = (cos\, t\, , sin\, t)$ as they satisfy the $|U|^2 = 1$, we have the following simplified result:

$$R\{f(x,y)\}(P) = \int_{-\infty}^{\infty} f(P\, cos\, t - k\, sin\, t\, , P\, sin\, t + k\, cos\, t)\, dk$$

<u>Example 1.1:</u> Calculate $R\{K(x,y)\}(P)$ assuming $K(x,y) = exp(-x^2 - y^2)$.

By making use of the above obtained result, we have:

$$R\{K(x,y)\}(P) = \int_{-\infty}^{\infty} exp(-(P\, cos\, t - k\, sin\, t)^2 + (\, P\, sin\, t + k\, cos\, t)^2)\, dk$$

$$R\{K(x,y)\}(P) = \int_{-\infty}^{\infty} exp\big(-k^2(sin^2 t + cos^2 t) - P^2(sin^2 t + cos^2 t)\big)\, dk$$

$$R\{K(x,y)\}(P) = exp(-P^2)\int_{-\infty}^{\infty} exp(-k^2)\, dk = \boldsymbol{exp}(-\boldsymbol{P}^2)\sqrt{\boldsymbol{\pi}}$$

Similarly, we can extend the original definition of Radon transforms to n-coordinates. Let us define, $X = [x_i]$ and $U = [u_i]\ \forall i \leq n$ such that $U \cdot X = U^T X = \sum_{i\leq n} u_i x_i$. Also, assume the constraint line $L:\ U^T X = P$, all X that satisfy L fall under the region S^n and m being the specialized distribution function.

From the first principles, we have:

$$R\{f(X)\}(P,U) = \int_L f(X)\, dm(X) = \int_{X \in S^n} f(X)\, dm(X)$$

Once again, we will be using the Dirac-delta δ function. Realize that, $\delta(P - U^T X)$ perfectly satisfies the constraint line L. Hence, the generalized Radon transform is given as:

$$R\{f(X)\}(P,U) = \int_{X \in \mathbb{R}^n} f(X)\delta(P - U^T X)\, dX = F(P,U)$$

This definition will be fundamental to all the upcoming sections.

Important properties of Radon transforms

By using the definition of Radon transforms, it becomes quite easy to prove few basic yet important identities. Let us assume, that $R\{k(X)\}(P,U) = K(P,U)$ and $R\{g(X)\}(P,U) = G(P,U)$, where all other prescribed assumptions hold (i.e. $X = [x_i]$ and $U = [u_i]\ \forall i \leq n$).

Linearity:

$$R\{k+g\}(P,U) = \int_{X \in \mathbb{R}^n} \big(k(X) + g(X)\big)\delta(P - U^T X)\, dX$$

$$R\{k+g\}(P,U) = \int_{X \in \mathbb{R}^n} k(X)\delta(P - U^T X)\, dX + \int_{X \in \mathbb{R}^n} g(X)\delta(P - U^T X)\, dX$$

$$R\{k+g\}(P,U) = R\{k\}(P,U) + R\{g\}(P,U)$$

This proves that Radon transforms are linear by nature.

Homogeneity {let c be any real number):

$$R\{ck\}(P,U) = \int_{X \in \mathbb{R}^n} ck(X)\delta(P - U^T X)\, dX = c\int_{X \in \mathbb{R}^n} k(X)\delta(P - U^T X)\, dX$$

$$R\{ck\}(P,U) = cR\{k\}(P,U)$$

Shifting (let a be any real number):

$$R\{k(x+a,y+a)\}(P,t) = \int_{x,y\in\mathbb{R}^2} k(x+a,y+a)\delta(P - x\cos t - y\sin t)dx\,dy$$

On assuming, $x' = x + a, dx' = dx$ and $y' = y + a, dy' = dy$ we have:

$$= \int_{x',y'\in\mathbb{R}^2} k(x',y')\delta(P + a\cos t + a\sin t - x'\cos t - y'\sin t)dx'\,dy'$$

$$R\{k(x+a,y+a)\}(P,t) = R\{k(x,y)\}(P + a\cos t + a\sin t\,, t)$$

Scaling (Let A to be a symmetric, non-singular real $n \times n$ matrix, such that $det(A) = 1$):

$$R\{k(AX)\}(P,U) = \int_{X\in\mathbb{R}^n} k(AX)\delta(P - U^T X)\,dX$$

Let, $Y = AX$ and as a consequence $dY = dX$ (Note that: $B = A^{-1}$ and in this case $B^T = B = A^{-1}$):

$$= \int_{Y\in\mathbb{R}^n} k(Y)\delta\big(P - U^T(BY)\big)\,dY = \int_{Y\in\mathbb{R}^n} k(Y)\delta(P - (B^T U)^T Y)\,dY$$

$$R\{k(AX)\}(P,U) = K(P, A^{-1}U)$$

<u>Example 2.1:</u> Assume $R\{m(x,y,z)\}(P,u,v,w) = M(P,u,v,w)$. Based on this, compute the following:

(i) $R\{m(x+a,y+b,z+c)\}(P,u,v,w)$ (where, a,b,c are real)

(ii) $R\{m(ax,by,cz)\}(P,u,v,w)$ (where, $a,b,c \neq 0$ and are positive.)

By letting $l = x + a$, $s = y + b$ and $k = z + c$ and as a consequence we have $dl = dx$, $ds = dy$ and $dk = dz$:

$$= \iiint_{l,s,k\,\in\,\mathbb{R}^3} m(l,s,k)\delta\big(P - u(l-a) - v(s-b) - w(k-c)\big)\,dl\,ds\,dk$$

$$= \iiint_{l,s,k\,\in\,\mathbb{R}^3} m(l,s,k)\delta(P + ua + vb + wc - ul - vs - wk)\,dl\,ds\,dk$$

Therefore, we obtain the following result for (i):

$$= M(P + ua + vb + wc, u, v, w)$$

For the second part, let $l = ax$, $s = by$ and $k = cz$ and as a consequence we have $\frac{dl}{a} = dx$, $\frac{ds}{b} = dy$ and $\frac{dk}{c} = dz$:

$$= \iiint_{l,s,k \in \mathbb{R}^3} m(l,s,k)\delta\left(P - u\left(\frac{l}{a}\right) - v\left(\frac{s}{b}\right) - w\left(\frac{k}{c}\right)\right)\left(\frac{dl}{a}\right)\left(\frac{ds}{b}\right)\left(\frac{dk}{c}\right)$$

$$= \iiint_{l,s,k \in \mathbb{R}^3} m(l,s,k)\delta\left(P - \frac{u}{a}(l) - \frac{v}{b}(s) - \frac{w}{c}(k)\right)\left(\frac{dl}{a}\right)\left(\frac{ds}{b}\right)\left(\frac{dk}{c}\right)$$

Therefore, we obtain the following result for (ii):

$$R\{m(ax, by, cz)\}(P, u, v, w) = \left(\frac{1}{abc}\right) M\left(P, \frac{u}{a}, \frac{v}{b}, \frac{w}{c}\right)$$

Shifting in transform parameters (let $A = [a_i]\ \forall i \le n$. Meaning, $dim(A) = 1 \times n$):

$$R\{k(X)\}(P + U^T A, U) = \int_{X \in \mathbb{R}^n} k(X)\delta(P + U^T A - U^T X)\, dX$$

$$= \int_{X \in \mathbb{R}^n} k(X)\delta\big(P - U^T(X - A)\big)\, dx = \int_{X \in \mathbb{R}^n} k(X + A)\delta(P - U^T X)\, dX$$

$$R\{k(X)\}(P + U^T A, U) = R\{k(X + A)\}(P, U)$$

Piece-wise continuity:

Assume that the function $m(X)$ can be described as follows:

$$m(X) = \begin{cases} k(X); X \in A^n \\ g(X); X \notin A^n \end{cases}$$

Where, A^n is a n-sphere with some positive radius. Important fact to note, union of $X \in A^n$ and $X \notin A^n$ results in $\mathbb{R}^n$ (i.e. $A^n \cup (\mathbb{R}^n - A^n) = \mathbb{R}^n$), which helps us in determining the Radon transform of $m(X)$:

$$R\{m(X)\}(P, U) = \int_{X \in \mathbb{R}^n} m(X)\delta(P - U^T X)\, dX$$

$$= \int_{X \in A^n} k(X)\delta(P - U^T X)\, dX + \int_{X \notin A^n} g(X)\delta(P - U^T X)\, dX$$

This covers all the essential basic identities of Radon transforms. Next up, we will discuss its connections with other transforms and convolution in Radon transforms:

Fundamental analysis of Radon transforms

Let us quickly familiarize ourselves with the definition of a Fourier transform. There are many variations across fields and all of these definitions are by and large equivalent in nature but the one we will be using is as follows:

$$F\{f(x)\}(k) = \int_{x\in\mathbb{R}} e^{-ikx} f(x)\, dx = \hat{f}(k)$$

The above definition of Fourier transform can be scaled to n-dimensions by assuming, $K = [k_i], X = [x_i]\ \forall i \leq n$.

$$F\{f(X)\}(K) = \int_{X\in\mathbb{R}^n} e^{-iK^T X} f(X)\, dX = \hat{f}(K)$$

Now, let us set $K = sU$. Where, U is an n-dimensional vector $U = [u_i]\ \forall i \leq n$. Note that s is a scalar in this case.

$$\hat{f}(sU) = \int_{X\in\mathbb{R}^n} e^{-isU^T X} f(X)\, dX$$

Now comes the most important identity of this derivation. From the basic principles of Dirac-delta δ function operating under integral sign, we realize the following:

$$\int_{y\in\mathbb{R}} e^{-isy} \delta(y - U^T X)\, dy = e^{-isU^T X}$$

Incorporating this fact in the derivation assists in rewriting the previous expression as such:

$$\hat{f}(sU) = \int_{X\in\mathbb{R}^n} f(X) \int_{y\in\mathbb{R}} e^{-isy} \delta(y - U^T X)\, dy\, dX$$

Interchanging the integral signs following from the basic assumption that $F\{f\}$ converges to a real finite quantity.

$$\hat{f}(sU) = \int_{y\in\mathbb{R}} e^{-is} \int_{X\in\mathbb{R}^n} f(X)\delta(y - U^T X)\,dX\,dy$$

Realize that the inner integral is simply the Radon transform of f with respect to (y, U). Assume that, $R\{f(x)\}(y, U) = f^*(y, U)$.

$$\hat{f}(sU) = \int_{y\in\mathbb{R}} e^{-isy} f^*(y, U)\,dy$$

The ultimate integral is simply the Fourier transform of $f^*(y, u)$ taken with respect to s. Hence, our final result being:

$$\hat{f}(sU) = F_y\{f^*(y, U)\}(s)$$

The following result is also known as **Fourier slice theorem**:

$$F\{f\}(sU) = F_y\{R\{f\}(y, U)\}(s)$$

Similarly, there is a direct correlation between double-sided Laplace transform and Radon transform. We define a double-sided Laplace transform as follows:

$$L\{f(x)\}(p) = \int_{x\in\mathbb{R}} e^{-px} f(x)\,dx = \tilde{f}(p)$$

For the upcoming result we consider the usual argument $X = [x_i]\ \forall i \le n$ and the function transformed being f.

$$L\{f(X)\}(sP) = \int_{X\in\mathbb{R}^n} e^{-sP^T X} f(X)\,dX$$

Where, s is a scalar and $P = [p_i]\ \forall i \le n$. Also, assuming that $L\{f(x)\}(sP) = \tilde{f}(sP)$.

$$\tilde{f}(sP) = \int_{X\in\mathbb{R}^n} f(X) \int_{y\in\mathbb{R}} e^{-sy}\delta(y - P^T X)\,dy\,dX$$

$$\tilde{f}(sP) = \int_{y\in\mathbb{R}} e^{-s} \int_{X\in\mathbb{R}^n} f(X)\delta(y - P^T X)\, dX\, dy$$

Assuming, $R\{f(x)\}(y, P) = f^*(y, P)$ we have:

$$\tilde{f}(sP) = \int_{y\in\mathbb{R}} e^{-sy} f^*(y, P)\, dy = L_y\{f^*(y, P)\}(s)$$

This can be alternatively stated as such:

$$L\{f\}(sP) = L_y\{R\{f\}(y, P)\}(s)$$

We will now be discussing the **convolution theorem** in Radon transforms.

Assume functions being transformed as $m(X)$ and $n(X)$ such that $X = [x_i]\ \forall i \leq n$. Recall that the Fourier convolution is given as such:

$$m * n = \int_{y\in\mathbb{R}} m(y)n(X - y)\, dy$$

Using the Fourier convolution with m, n as an argument for the radon transform, gives us the following:

$$R\{m * n\}(P, U) = R\left\{\int_{y\in\mathbb{R}} m(y)n(X - y)\, dy\right\}(P, U)$$

$$R\{m * n\}(P, U) = \int_{X\in\mathbb{R}^n} \delta(P - U^T X) \int_{y\in\mathbb{R}} m(y)n(X - y)\, dy\, dX$$

By interchanging the integral signs, we simplify further.

$$R\{m * n\}(P, U) = \int_{y\in\mathbb{R}} m(y) \int_{X\in\mathbb{R}^n} n(X - y)\delta(P - U^T X)\, dX\, dy$$

Let, $\begin{matrix} V = X - y \\ dV = dX \end{matrix}$:

$$R\{m*n\}(P,U) = \int_{y\in\mathbb{R}} m(y) \int_{V\in\mathbb{R}^n} n(V)\delta(P - U^T y - U^T V)\, dV\, dy$$

Now, we will assume that $R\{n\}(P,U) = N(P,U)$ and $R\{m\}(P,U) = M(P,U)$.

$$R\{m*n\}(P,U) = \int_{y\in\mathbb{R}} m(y)N(P - U^T y, U)\, dy$$

By using the basic Dirac-delta δ function distribution property, we can rewrite the above statement.

$$R\{m*n\}(P,U) = \int_{y\in\mathbb{R}} m(y) \int_{s\in\mathbb{R}} N(P - s, U)\delta(s - U^T y)\, ds\, dy$$

$$R\{m*n\}(P,U) = \int_{S\in\mathbb{R}} N(P - s, U) \int_{y\in\mathbb{R}} m(y)\delta(s - U^T y)\, dy\, ds$$

Recognize that the inner integral is simply the Radon transform of m w.r.t (s,U), which gives us the following.

$$R\{m*n\}(P,U) = \int_{S\in\mathbb{R}} N(P - s, U)M(s,U)\, ds = N(P,U) * M(P,U)$$

Ultimately, our final result can be stated as follows:

$$R\{m*n\}(P,U) = R\{m\}(P,U) * R\{n\}(P,U)$$

The same can also be proved using the Fourier slice theorem. Let us firstly restate the Fourier slice theorem using Fourier convolution between m, n.

$$F\{n(X) * m(X)\}(sU) = F_P\{R\{n(X) * m(X)\}(P,U)\}(s)$$

Taking F_P^{-1} on both sides of the equation, leaves us the following:

$$R\{n(X) * m(X)\}(P,U) = F_P^{-1}\{F\{n(X) * m(X)\}(sU)\}(P)$$

$$R\{n(X) * m(X)\}(p,U) = F_P^{-1}\{F\{n(X)\}(sU)F\{m(X)\}(sU)\}(P)$$

Here, assume that $F\{n(X)\}(sU) = \hat{n}(sU)$ and $F\{m(X)\}(sU) = \hat{m}(sU)$.

$$R\{n(X) * m(X)\}(p, U) = F_P^{-1}\{\hat{n}(sU)\hat{m}(sU)\}(P)$$

$$R\{n(X) * m(X)\}(p, U) = F_P^{-1}\{\hat{n}(sU)\}(P) * F_p^{-1}\{\hat{m}(sU)\}(P)$$

Realize that, $F_P^{-1}\{\hat{n}(sU)\}(P) = R\{n\}(P, U)$ and $F_P^{-1}\{\hat{m}(sU)\}(P) = R\{m\}(P, U)$, by which we obtain the final result:

$$R\{n * m\}(P, U) = R\{n\}(P, U) * R\{m\}(P, U)$$

Computations in Radon transforms

Let us assume the two-dimensional function $h(x, y)$ which is being transformed over the constraint coefficients u, v and principal constant P. The question that we raise here is, how do we compute the Radon transform of $\frac{\partial h(x,y)}{\partial x}$. To answer the same, let us rewrite the required transform as such:

$$R\left\{\frac{\partial h(x, y)}{\partial x}\right\}(P, u, v) = R\left\{\frac{h\left(x + \frac{t}{u}, y\right) - h(x, y)}{t\,/\,u}\right\}(P, u, v)$$

Here we set $\frac{t}{u} \to 0$. Now, let us quickly compute the Radon transform of the function $h\left(x + \frac{t}{u}, y\right)$.

$$R\left\{h\left(x + \frac{t}{u}, y\right)\right\}(P, u, v) = \int\limits_{x,y\in\mathbb{R}^2} h\left(x + \frac{t}{u}, y\right)\delta(P - ux - vy)\,dx\,dy$$

Let, $\begin{matrix} w = x + \frac{t}{u} \\ dw = dx \end{matrix}$:

$$R\left\{h\left(x + \frac{t}{u}, y\right)\right\}(P, u, v) = \int\limits_{w,y\in\mathbb{R}^2} h(w, y)\delta\left(P - u\left(w - \frac{t}{u}\right) - vy\right)dw\,dy$$

$$= \int\limits_{w,y\in\mathbb{R}^2} h(w, y)\delta(P + t - uw - vy)\,dw\,dy = R\{h(x, y)\}(P + t, u, v)$$

We now apply this result to our initial expression, which leads to the following:

$$R\left\{\frac{\partial h(x, y)}{\partial x}\right\}(P, u, v) = \frac{u}{t}\{R\{h(x, y)\}(P + t, u, v) - R\{h(x, y)\}(P, u, v)\}$$

$$R\left\{\frac{\partial h(x,y)}{\partial x}\right\}(P,u,v) = u\frac{\partial}{\partial P}R\{h(x,y)\}(P,u,v)$$

A similar analogous result is observed in case of $\frac{\partial h}{\partial y}$:

$$R\left\{\frac{\partial h(x,y)}{\partial y}\right\}(P,u,v) = v\frac{\partial}{\partial P}R\{h(x,y)\}(P,u,v)$$

Similarly, the derivative computations in the case of two-dimensional function $h(x,y)$ can be generalized as follows (where $m,n \in \mathbb{Z}_{>0}$):

$$R\left\{\frac{\partial^{(m+n)}h(x,y)}{\partial x^m \partial y^n}\right\}(P,u,v) = u^m v^n \frac{\partial^{(m+n)}}{\partial P^{(m+n)}}R\{h(x,y)\}(P,u,v)$$

Another nice way to compute the Radon transform of $\frac{\partial h}{\partial x}$ is by using the integration by parts rule.

$$R\left\{\frac{\partial h(x,y)}{\partial x}\right\}(P,u,v) = \int\limits_{y\in\mathbb{R}} \int\limits_{x\in\mathbb{R}} \frac{\partial h}{\partial x}\delta(P-ux-vy)\,dx\,dy$$

$$= \int\limits_{y\in\mathbb{R}} \left[\delta(p-ux-vy)h(x,y)|_{x\in\mathbb{R}} - \int\limits_{x\in\mathbb{R}} \frac{\partial}{\partial x}\big(\delta(p-ux-vy)\big)h(x,y)dx\right]dy$$

It is important to note that, $\frac{\partial}{\partial x}\big(\delta(P-ux-vy)\big) = -u\frac{\partial}{\partial P}\big(\delta(P-ux-vy)\big)$.

$$= u\frac{\partial}{\partial P}\iint\limits_{x,y\in\mathbb{R}^2} h(x,y)\,d(P-ux-vy)\,dx\,dy = u\frac{\partial}{\partial P}R\{h(x,y)\}(P,u,v)$$

Now, let us consider the n-dimenstional case by using the function $h(X)$, where $X = [x_i]\ \forall i \leq n$ and the constraint vector $U = [u_i]\ \forall i \leq n$. We are tasked to find the Radon transform of $\frac{\partial h(X)}{\partial x_l}$ (such that, $1 \leq l \leq n$). Firstly, rewriting the partial derivative similar to our previous exposition.

$$R\left\{\frac{\partial h(X)}{\partial x_l}\right\}(P,U) = R\left\{\frac{h\left(x_1, x_2, \dots, x_l + \frac{t}{u_l}, \dots, x_n\right) - h(X)}{t\ /\ u_l}\right\}$$

Now, by using the shifting property of Radon transforms, we can rewrite the above expression.

$$R\left\{\frac{\partial h(X)}{\partial x_l}\right\}(P,U) = \frac{R\{h(x)\}(P,u_1,u_2,\dots,u_l+t,\dots,u_n) - R\{h(X)\}(P,U)}{t/u_l}$$

$$R\left\{\frac{\partial h(X)}{\partial x_l}\right\}(P,U) = u_l\frac{\partial}{\partial P}R\{h(X)\}(P,U)$$

And similarly, the following generalized derivative can be transformed as such (given, $m \in \mathbb{Z}_{>0}$):

$$R\left\{\frac{\partial^m h(x)}{\partial x_l^m}\right\}(P,U) = u_l^m\frac{\partial^m}{\partial P^m}\left(R\{h(X)\}(P,U)\right)$$

By using this identity, the following property can be derived which proves to be pivotal in simplifying many n-dimensional PDEs. Let us try to find the Radon transform of $\nabla h(X)$.

$$R\{\nabla h(X)\}(P,U) = R\left\{\sum_{1\le i\le n}\frac{\partial^2}{\partial x_i^2}h(x)\right\}(P,U)$$

By using the property of Linearity, we have the following:

$$R\{\nabla h(X)\}(P,U) = \sum_{1\le i\le n}R\left\{\frac{\partial^2}{\partial x_i^2}h(X)\right\}(P,U)$$

On using the above derived property, our expression simplifies as such:

$$R\{\nabla h(X)\}(P,u) = \sum_{1\le i\le n}u_i^2\frac{\partial^2}{\partial P^2}R\{h(X)\}(P,u) = \frac{\partial^2}{\partial P^2}R\{h(X)\}(P,u)\sum_{1\le i\le n}u_i^2$$

As we already know that $|U| = 1$, we can confidently state the following result.

$$R\{\nabla h(X)\}(P,U) = \frac{\partial^2}{\partial P^2}R\{h(X)\}(P,U)$$

This property is quite helpful in certain PDEs as it converts an n-dimensional PDE into a two-dimensional PDE.

With that done, we can now explore few important computations centred around Hermite polynomials, $H_n(x)$. Firstly, a Hermite polynomial is defined as such using the Rodrigues' formula:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\left(e^{-x^2}\right)$$

By using this formulation, let us try to compute the Radon transform of $e^{-x^2-y^2} H_n(x) H_m(y)$. Firstly, rewrite the argument using the Rodrigues' formula.

$$R\left\{e^{-x^2-y^2} H_n(x) H_m(y)\right\}(P,U) = R\left\{(-1)^{n+m} D_x^n\left(e^{-x^2}\right) D_y^m\left(e^{-y^2}\right)\right\}(P,U)$$

Now, on using the Integral definition of Radon transforms, we have the following:

$$= \int\limits_{y\in\mathbb{R}} \int\limits_{x\in\mathbb{R}} (-1)^{n+m} D_x^n\left(e^{-x^2}\right) D_y^m\left(e^{-y^2}\right) \delta(P - ux - vy)\, dx\, dy$$

After applying Integration by parts n-number of times w.r.t x and m-number of times w.r.t y, we obtain the following:

$$= (-1)^{n+m} u^n v^m D_P^{m+n} \int\limits_{y\in\mathbb{R}} \int\limits_{x\in\mathbb{R}} e^{-x^2-y^2} \delta(P - ux - vy)\, dx\, dy$$

$$= \sqrt{\pi}(-1)^{n+m}(u^n)(v^m) D_P^{m+n}\left(e^{-P^2}\right)$$

$$= \sqrt{\pi}(u^n)(v^m) e^{-P^2}\left((-1)^{n+m} e^{P^2} D_P^{m+n}\left(e^{-P^2}\right)\right)$$

Therefore, our final result is stated as such by assuming $u = cos\, t$ and $v = sin\, t$:

$$R\left\{H_n(x) H_m(y) e^{-x^2-y^2}\right\}(P,t) = \sqrt{\pi}(cos\, t)^n (sin\, t)^m e^{-P^2} H_{n+m}(P)$$

<u>Example 4.1:</u> Compute the Radon transform of $xye^{-x^2-y^2}$ with respect to (P, u, v), where you can later assume $u = cos\, t$ and $v = sin\, t$.

Firstly, recognize that the term xy can be written as a product of two Hermite polynomials.

$$H_1(x) = (-1) e^{x^2} \frac{d}{dx}\left(e^{-x^2}\right) = -e^{x^2} e^{-x^2}(-2x) = 2x$$

$$H_1(y) = (-1)e^{y^2}\frac{d}{dy}\left(e^{-y^2}\right) = -e^{y^2}e^{-y^2}(-2y) = 2y$$

Therefore, $xy = \frac{H_1(x)H_1(y)}{4}$. By using this identity, we have:

$$R\left\{xye^{-x^2-y^2}\right\}(P,u,v) = \frac{1}{4}R\left\{H_1(x)H_1(y)e^{-x^2-y^2}\right\}(P,u,v)$$

Now, we make use of the earlier derived result of the Radon transform of generalized Hermite polynomials in two-dimensions.

$$R\left\{xye^{-x^2-y^2}\right\}(P,cos\,t\,,sin\,t) = \frac{\sqrt{\pi}}{4}(cos\,t\,sin\,t)e^{-P^2}H_2(P)$$

Here, $H_2(P) = 4P^2 - 2$. Ultimately, we have the following result:

$$R\left\{xye^{-x^2-y^2}\right\}(P,cos\,t\,,sin\,t) = \sqrt{\pi}(cos\,t\,sin\,t)e^{-P^2}\left(P^2 - \frac{1}{2}\right)$$

With that done, we can now discuss a bit more regarding the possible computations using the properties of Kronecker delta-δ function.

Consider some function $f(x+y)$, such that it is differentiable in the $\mathbb{R}^2$. If we were to assume $E = x + y$, then it becomes quite trivial to observe the following result:

$$\frac{d}{dx}\left(f(x+y)\right) = \frac{d}{dy}\left(f(x+y)\right) = f'(E)$$

Similarly, if we consider the Kronecker delta-δ function, we observe a similar identity.

$$\frac{d}{dx}\left(\delta(x+y)\right) = \frac{d}{dy}\left(\delta(x+y)\right) = \delta'(E)$$

Now, if we assume $E = x - y$, then the identity gets tweaked as follows:

$$\frac{d}{dx}\left(\delta(x-y)\right) = -\frac{d}{dy}\left(\delta(x-y)\right)$$

Consequently, for $E = x - ay$, we have:

$$\frac{d}{dx}\left(\delta(x-ay)\right) = \frac{-1}{a}\frac{d}{dy}\left(\delta(x-ay)\right)$$

For the next result, we assume $E = x - A^T Y$, such that $Y = [y_i]$ & $A = [a_i]$ $\forall i \leq n$. Let, $1 \leq k \leq n$.

$$\frac{d}{dx}\left(\delta(x - A^T Y)\right) = \frac{-1}{a_k}\frac{d}{dy_k}\left(\delta(x - A^T Y)\right)$$

Ultimately, the general result is observed for $E = X - A^T Y$, such that $Y = [y_i], A = [a_i]$ & $X = [x_i]$ $\forall i \leq n$. Let, $1 \leq k \leq n$.

$$\frac{d}{dx_k}\left(\delta(X - A^T Y)\right) = \frac{-1}{a_k}\frac{d}{dy_k}\left(\delta(X - A^T Y)\right)$$

This property becomes particularly useful in evaluating and simplifying derivatives of Radon transforms.

For instance, let us assume $R\{f(X)\}(P, U) = F(P, U)$ such that $X = [x_i]$ $\forall i \leq n$. Let, $1 \leq l, m \leq n$ & $k \in \mathbb{Z}_{>0}$.

$$\frac{\partial F(P, U)}{\partial u_l} = \int\limits_{X \in \mathbb{R}^n} f(X)\frac{d}{du_l}\left(\delta(P - U^T X)\right) dX$$

$$= -\int\limits_{X \in \mathbb{R}^n} x_l f(X)\frac{d}{dP}\left(\delta(P - U^T X)\right) dX = -\frac{\partial}{\partial P}R\{x_l f(X)\}(P, U)$$

We can also extend the above result as follows:

$$\frac{\partial^2 F(P, U)}{\partial u_l \partial u_m} = \int\limits_{X \in \mathbb{R}^n} f(X)\frac{\partial^2}{\partial u_l \partial u_m}\left(\delta(P - U^T X)\right) dX$$

$$= \int\limits_{X \in \mathbb{R}^n} x_l x_m f(X)\frac{\partial^2}{\partial P^2}\left(\delta(P - U^T X)\right) dX = \frac{\partial^2}{\partial P^2}R\{x_l x_m f(X)\}(P, U)$$

Therefore, the generalized result can be stated as such:

$$\frac{\partial^{(k)} F(P, U)}{\partial u_l^{(k)}} = (-1)^k \frac{\partial^k}{\partial P^k}R\left\{x_l^k f(X)\right\}(P, U)$$

Now, we will explore disk-based computations in Radon transforms. So far, we have evaluated functions on the entire real n-sphere but Radon transform's usefulness extends to finite intervals as well. To discuss that, we will consider the following function:

$$f(x,y) = \begin{cases} A\ ;\ x^2 + y^2 \leq a \\ 0\ ;\ x^2 + y^2 > a \end{cases}$$

Where, A is a constant and $a > 0$.

To calculate the Radon transform of $f(x,y)$, let us first represent it using the standard Kronecker delta notation. Here, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $U = \begin{bmatrix} u \\ v \end{bmatrix}$.

$$R\{f(X)\}(P,U) = \int\limits_{X \in \mathbb{R}^2} f(X)\delta(P - U^T X)\, dX$$

As per usual, we define $B = \begin{bmatrix} u & -v \\ v & u \end{bmatrix}$ and perform a substitution. Let, $X = BY$ and $dX = |det(B)|\, dY = dY$ such that $Y = \begin{bmatrix} m \\ n \end{bmatrix}$.

$$R\{f(X)\}(P,U) = \int\limits_{Y \in \mathbb{R}^2} f(BY)\delta\big(P - U^T(BY)\big)\, dY$$

Here, $U^T(BY) = [u\ v] \begin{bmatrix} u & -v \\ v & u \end{bmatrix} \begin{bmatrix} m \\ n \end{bmatrix} = [u\ v] \begin{bmatrix} mu - nv \\ mv + nu \end{bmatrix} = mu^2 + mv^2 = m$.

$$R\{f(x,y)\}(P,u,v) = \int\limits_{m,n \in \mathbb{R}^2} f(mu - nv, mv + nu)\delta(P - m)\, dm\, dn$$

$$R\{f(x,y)\}(P,u,v) = \int\limits_{n \in \mathbb{R}} f(Pu - nv, Pv + nu)\, dn$$

Now, it becomes important to realize the identity $P^2 + n^2 = x^2 + y^2$. From the definition of $f(x,y)$, the function has a non-zero value only when $x^2 + y^2 \leq a$. Therefore, the above identity now turns into $n^2 \leq a - P^2$ which implies the integral will have a non-zero value only if evaluated from $n \in \left[-\sqrt{a - P^2}, \sqrt{a - P^2}\right]$.

$$R\left\{f(x,y)|_{x^2 + y^2 \leq a}\right\}(P,u,v) = \int\limits_{-\sqrt{a - P^2}}^{\sqrt{a + P^2}} f(Pu - nv, Pv + nu)\, dn$$

$$R\{f(x,y)|_{x^2+y^2\le a}\}(P,u,v) = \int_{-\sqrt{a-P^2}}^{\sqrt{a+P^2}} A\,dn = 2A\sqrt{a-P^2}$$

As already mentioned, in the case of $x^2 + y^2 > a$ the function $f(x,y)$ results as 0, which implies the following:

$$R\{f(x,y)|_{x^2+y^2>a}\}(P,u,v) = 0$$

Therefore, the Radon transform of the function $f(x,y)$ can be written as follows:

$$R\{f(x,y)\}(P,u,v) = \begin{cases} 2A\sqrt{a-P^2}\,; |P| \le \sqrt{a} \\ 0\,;\ |P| > \sqrt{a} \end{cases}$$

Before moving further, let us discuss one very important general property that captures various compressed parameters. Suppose, we are tasked with finding the Radon transform of $k\left(\frac{x}{a},\frac{y}{b}\right)$ (where $a, b \neq 0$) with respect to transform parameters (P,u,v), assuming $R\{k(x,y)\}(P,u,v) = K(P,u,v)$.

$$R\left\{k\left(\frac{x}{a},\frac{y}{b}\right)\right\}(P,u,v) = \int_{x,y\in\mathbb{R}^2} k\left(\frac{x}{a},\frac{y}{b}\right)\delta(P-ux-vy)\,dx\,dy$$

Let, $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$, $U = \begin{bmatrix} u \\ v \end{bmatrix}$ and $Y = \begin{bmatrix} m \\ n \end{bmatrix}$. So, we set $Y = B^{-1}X$ and as a consequence we have $dY = |det(B^{-1})|\,dX = dX/|ab|$.

$$R\{K(B^{-1}X)\}(P,U) = |ab| \int_{Y\in\mathbb{R}^2} K(Y)\delta(P - U^TBY)\,dY$$

Meaning, the above expression can be stated as such (such that, $l = \sqrt{(au)^2 + (bv)^2}$).

$$R\left\{k\left(\frac{x}{a},\frac{y}{b}\right)\right\}(P,u,v) = |ab|K\left(P,\frac{alu}{l},\frac{blv}{l}\right)$$

As, $M = \left(\frac{au}{l},\frac{bv}{l}\right)$ is a unit vector, we can confidently assert the following:

$$R\left\{k\left(\frac{x}{a},\frac{y}{b}\right)\right\}(P,U) = |ab|K(P,lM)$$

By following properties of Radon transforms and realizing $U = \begin{bmatrix} cos\,t \\ sin\,t \end{bmatrix}$, we have:

$$R\left\{k\left(\frac{x}{a},\frac{y}{b}\right)\right\}(P,t)=\frac{|ab|}{l}K\left(\frac{P}{l},M\right)$$

Such that, $l=\sqrt{(a\cos t)^2+(b\sin t)^2)}$.

Thanks to this amazing property, we can calculate the Radon transform of $exp\left(-\left(\frac{x}{a}\right)^2-\left(\frac{y}{b}\right)^2\right)$ with ease. Firstly, recall that $R\{exp(-x^2-y^2)\}(P,u,v)=\sqrt{\pi}e^{-P^2}$. Now, by using the produced property, we have the following.

$$R\left\{exp\left(-\left(\frac{x}{a}\right)^2-\left(\frac{y}{b}\right)^2\right)\right\}(P,t)=\frac{\sqrt{\pi}|ab|}{l}e^{-\left(\frac{P}{l}\right)^2}$$

Extending our curiosity further, we will now discuss the case of an elliptical disk. Let us consider the function $h(x,y)$ to be defined as follows:

$$h(x,y)=\begin{cases}A\,;\ \left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2\le L\\ 0\,;\ \left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2>L\end{cases}$$

Where, A is some real constant. As per usual, considering P as Principal parameter and $U=(u,v)$ as the weight parameters.

By making use of the previously derived property, we can state the following confidently:

$$R\left\{h(x,y)|_{\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2\le L}\right\}(P,U)=\frac{|ab|}{l}R\left\{h(x,y)|_{x^2+y^2\le L}\right\}\left(\frac{P}{l},M\right)$$

From our previous exposition, the above expression has the following simplification:

$$R\left\{h(x,y)|_{\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2\le L}\right\}(P,U)=\frac{2|ab|}{l}\left(A\sqrt{L-\left(\frac{P}{l}\right)^2}\right)$$

Therefore, the final result of the Radon transform of an elliptic disk is asserted as such:

$$R\{h(x,y)\}(P,U) = \begin{cases} \frac{2|ab|}{l}\left(A\sqrt{L-\left(\frac{P}{l}\right)^2}\right) ; \frac{|P|}{l} \le \sqrt{L} \\ 0 ; \frac{|P|}{l} > \sqrt{L} \end{cases}$$

Lastly, we will be discussing the Radon transform of delta functions. Firstly, let us assume a two-dimensional case where we are tasked to find the Radon transform of $\delta(x-a)\delta(y-b)$ (Note that, $\delta(x-a)\delta(y-b) = \delta(x-a, y-b)$ both are identical representations). Firstly, let us write down the transform using the integral definition:

$$R\{\delta(x-a,y-b)\}(P,t) = \int_{x,y\in\mathbb{R}^2} \delta(x-a,y-b)\delta(P-x\cos t-y\sin t)dx\,dy$$

Now realize that this integral will have a non-zero value only when $x=a, y=b$ and this is indeed the most important point of the given transform. So, we will ignore all other values of x, y except (a,b) and simplify the resulting expression:

$$R\{\delta(x-a,y-b)\}(P,t) = \delta(P-a\cos t-b\sin t) = \delta(P-P_0)$$

Here, $P_0 = a\cos t + b\sin t$.

Similarly, the same can be extended to n-dimensions. Let, $X = [x_i]$, $U = [u_i]$ & $A = [a_i]$ $\forall i \le n$. To calculate the Radon transform of $\delta(X-A)$:

$$R\{\delta(X-A)\}(P,U) = \int_{X\in\mathbb{R}^n} \delta(X-A)\delta(P-U^TX)\,dX$$

$$= \delta(P-U^TA) = \delta(P-P_0)$$

Here, $P_0 = U^TA = \sum_{i\le n} u_i a_i$.

Example 4.2: Compute the Radon transform of the disk characteristic function $\mu(r)$ with respect to P, described as follows:

$$\mu(r) = \begin{cases} 1 ; r^2 \le 1 \\ 0 ; r^2 > 1 \end{cases}$$

Here, r is radius described by the identity $r = \sqrt{x^2+y^2}$.

Firstly, recognize that the cylinder function $\mu(r)$ can be written using x, y and it will represent a unit-disk region.

$$\mu(x, y) = \begin{cases} 1\,;\ x^2 + y^2 \leq 1 \\ 0\,;\ x^2 + y^2 > 1 \end{cases}$$

Now, we will be taking the Radon transform of the above w.r.t P, t :

$$R\{\mu(x, y)\}(P, t) = \int_{n \in \mathbb{R}} \mu(P\, cos\, t - n\, sin\, t\,, P\, sin\, t + n\, cos\, t)\, dn$$

Now, recognize that in the case of $x^2 + y^2 \leq 1$, we use the identity $P^2 + n^2 = x^2 + y^2$, which leads to the inequality $n^2 \leq 1 - P^2$, which ultimately implies, the limits of the resulting integral are $\left[-\sqrt{1 - P^2}, \sqrt{1 - P^2}\right]$.

$$R\left\{\mu(x, y)|_{x^2+y^2\leq 1}\right\}(P, t) = \int_{-\sqrt{1-P^2}}^{\sqrt{1-P^2}} dn = 2\sqrt{1 - P^2}$$

On the contrary, for $x^2 + y^2 > 1$, we can clearly observe that $\mu(x, y) = 0$.

$$R\left\{\mu(x, y)|_{x^2+y^2>1}\right\}(P, t) = 0$$

Therefore, our final result can be stated as such:

$$R\{\mu(r)\}(P, t) = 2\mu(P)\sqrt{1 - P^2}$$

<u>Example 4.3:</u>Compute the Radon transform of the following function $h(x, y)$ with respect to (P, t)

$$h(x, y) = \begin{cases} \dfrac{1}{x^2 + y^2}\,;\ x^2 + y^2 \leq b \\ 0\,;\ x^2 + y^2 > b \end{cases}$$

Here, b is some positive real constant.

As per usual, we can write the Radon transform integral using standard trigonometric parameters:

$$R\{h(x, y)\}(P, t) = \int_{n \in \mathbb{R}} h(P\, cos\, t - n\, sin\, t\,, P\, sin\, t + n\, cos\, t)\, dn$$

We realize the identity $x^2 + y^2 = P^2 + n^2$ by which we can confidently state the inequality $n^2 \leq b - P^2$ which ultimately implies $n \in \left[-\sqrt{b - P^2}, \sqrt{b - P^2}\right]$.

$$R\left\{h(x,y)|_{x^2+y^2 \leq b}\right\}(P,t) = \int_{-\sqrt{b-P^2}}^{\sqrt{b-P^2}} \frac{dn}{P^2 + n^2} = \frac{1}{P} tan^{-1}\left(\frac{n}{P}\right)\Big|_{-\sqrt{b-P^2}}^{\sqrt{b-P^2}}$$

$$R\left\{h(x,y)|_{x^2+y^2 \leq b}\right\}(P,t) = \frac{2}{P} tan^{-1}\left(\frac{\sqrt{b - P^2}}{P}\right)$$

As $h(x,y)|_{x^2+y^2 > b} = 0$, we can state the following final result:

$$R\{h(x,y)\}(P,t) = \begin{cases} \frac{2}{P} tan^{-1}\left(\frac{\sqrt{b - P^2}}{P}\right); |P| \leq \sqrt{b} \\ 0\,;\ |P| > \sqrt{b} \end{cases}$$

Example 4.4:Consider the following function $N(x,y)$ which describes a square region of length $2b$. (Assume that b is a positive quantity).

$$N(x,y) = \begin{cases} 1\,; |x| \leq b \wedge |y| \leq b \\ 0\,; |x| > b \vee |y| > b \end{cases}$$

Compute the Radon transform of $N(x,y)$ with respect to transform parameters (P,t) based on the given conditions:

(i) At $t = 0$

(ii) At $t = \frac{\pi}{4}$

Firstly, let us write down the Radon transform of $N(x,y)$ in standard trigonometric notation:

$$R\{N(x,y)\}(P,t) = \int_{n \in \mathbb{R}} N(P\,cos\,t - n\,sin\,t\,, P\,sin\,t + n\,cos\,t)\,dn$$

Important constraints to keep in mind for this problem are, $|P\,cos\,t - n\,sin\,t| \leq b$ and $|P\,sin\,t + n\,cos\,t| \leq b$. Now, for the case of $t = 0$ the constraints simplify to $|P| \leq b$ and $|n| \leq b$. Thus, we have the following:

$$R\left\{N(x,y)|_{|x|,|y| \leq b}\right\}(P,t) = \int_{n=-b}^{b} dn = 2b$$

In the case of $|x| > b \vee |y| > b$, $N(x, y) = 0$. Thus, Radon transform of $N(x, y)$ at $t = 0$ is given by:

$$R\{N(x,y)\}(P,0) = \begin{cases} 2b\ ;\ |P| \le b \\ 0\ ;\ |P| > b \end{cases}$$

Now to calculate the Radon transform at $t = \frac{\pi}{4}$, we modify the constraints accordingly. $|P - n| \le \sqrt{2}b$ and $|P + n| \le \sqrt{2}b$ are the required constraints, squaring them both and adding them up, results in $P^2 + n^2 \le 2b$. Meaning, $n \in \left[-\sqrt{2b - P^2}, \sqrt{2b - P^2}\right]$.

$$R\left\{N(x,y)|_{|x|,|y|\le b}\right\}\left(P,\frac{\pi}{4}\right) = \int_{-\sqrt{2b-P^2}}^{\sqrt{2b-P^2}} dn = 2\sqrt{2b - P^2}$$

On the contrary, as $N(x, y) = 0$ when $|x| > b \vee |y| > b$. Therefore, our final result is:

$$R\{N(x,y)\}\left(P,\frac{\pi}{4}\right) = \begin{cases} 2\sqrt{2b - P^2}\ ; |P| \le b\sqrt{2} \\ 0\ ; |P| > b\sqrt{2} \end{cases}$$

With that done, we can now move forward to the next section and understand the machinery behind Inverse Radon transforms.

Inversion of Radon transforms

To perform a Radon inversion, we firstly need to recall the Fourier slice theorem given as such:

$$F\{f\}(sU) = F_y\{R\{f\}(y,U)\}(s)$$

Here, $sU = (s\cos t, s\sin t)$, $F\{f\}(sU) = \tilde{f}(sU)$, $R\{f\}(P,U) = \bar{f}(P,U)$. Important to note, we are only concerned with the two-dimensional case here.

$$\tilde{f}(sU) = F_y\left\{\bar{f}(y,U)\right\}(s)$$

Taking F^{-1} on both sides w.r.t sU, leads to the following:

$$f(x,y) = F^{-1}\left\{F_y\left\{\bar{f}(y,U)\right\}(s)\right\}$$

This can be represented as a double integral:

$$f(x,y) = \int\limits_{S\in\mathbb{R}} |s|\, ds \int\limits_{0}^{\pi} \tilde{f}(sU) e^{2\pi i s(x\cos t + y\sin t)}\, dt$$

Note that, here we are considering a slightly different version of Fourier transform but regardless of the kernel changes, all of the previously derived theorems and properties will hold.

Here, we have to set $U^T X = P$ which implies $x\cos t + y\sin t = P$.

$$f(x,y) = \int\limits_{S\in\mathbb{R}} |s|\, ds \int\limits_{0}^{\pi} e^{2i\pi sP} \tilde{f}(sU) dt$$

$$f(x,y) = \int\limits_{0}^{\pi} dt \int\limits_{S\in\mathbb{R}} e^{2i\pi sP} \tilde{f}(sU)|s|\, ds$$

The inner integral can be rewritten using Fourier convolution.

$$f(x,y) = \int\limits_{0}^{\pi} \left(F^{-1}\left\{\tilde{f}(sU)\right\} * F^{-1}\{|s|\}\right) dt$$

It is a well-established fact that $F^{-1}\left\{\tilde{f}(sU)\right\} = \bar{f}(P,U)$. Whereas, $F^{-1}\{|s|\}$ can be simplified as such:

$$F^{-1}\{|s|\} = F^{-1}\{s\, sgn(s)\} = F^{-1}\{s\} * F^{-1}\{sgn(s)\}$$

By making use of the Inverse Fourier transform table, the following is stated:

$$F^{-1}\{|s|\} = a_2 \delta'(P) * PV\left(\frac{1}{P}\right)$$

Here, $a_2 = \frac{1}{2\pi^2}$ and PV denotes Cauchy's Principal value of the given function. Returning back to our original expression:

$$f(x,y) = \frac{1}{2\pi^2} \int\limits_{0}^{\pi} \bar{f}(P,U) * \delta'(P) * PV\left(\frac{1}{P}\right) dt$$

Now we will tread the computations of $\bar{f}(P,U) * \delta'(P)$ to simplify the integral expression:

$$\bar{f}(P,U) * \delta'(P) = \frac{\partial \bar{f}(P,U)}{\partial P} * \delta(P) = \frac{\partial \bar{f}(P,U)}{\partial P}$$

This handy computation, results in the following simplified expression:

$$f(x,y) = \frac{1}{2\pi^2} \int_0^{\pi} \frac{\partial \bar{f}(P,U)}{\partial P} * PV\left(\frac{1}{P}\right) dt$$

Applying the Fourier convolution yet again. (Here, $\frac{\partial \bar{f}}{\partial P} = \bar{f}_P$):

$$f(x,y) = \frac{1}{2\pi^2} \int_{\theta}^{\pi} dt\, PV \int_{T\in\mathbb{R}} \frac{\bar{f}_T(T,U)}{p-T} dT$$

As we have already assumed $x\,cos\,t + y\,sin\,t = P$, we can substitute that back into the above expression. Also, we will be dropping PV from the expression similar to many of the older texts.

$$f(x,y) = -\frac{1}{2\pi^2} \int_0^{\pi} dt \int_{T\in R} \frac{\bar{f}_T(T,U)}{T - x\,cos\,t - y\,sin\,t} dT$$

By using the Hilbert transform definition, we have the following final result:

$$f(x,y) = -\frac{1}{2\pi} \int_0^{\pi} H\{\bar{f}_P(P,U) \big| x\,cos\,t + y\,sin\,t = P\} dt$$

That concludes the discussion for the Inverse Radon transform of two-dimensions. Now, we will be stepping up and work in three-dimensions instead. Similar assumptions are taken into account, $sU = (su, sv, sw)$, $f(X) = f(x,y,z)$, $F\{f\}(sU) = \tilde{f}(sU)$ and $R\{f\}(P,U) = \bar{f}(P,U)$. As we are in three-dimensional case now, we cannot use the parameter t (i.e. usage of polar coordinates) and instead have to assume $|U| = \sqrt{u^2+v^2+w^2} = 1$ as the condition for the inner integral.

$$f(x,y,z) = \int_{s\in\mathbb{R}^+} s^2\, ds \int_{|U|=1} e^{2\pi i s(U^T X)} \tilde{f}(sU)\, dU$$

As per usual, we set $U^T X = P$ leading to the following simplification:

$$f(x,y,z) = \int_{s\in\mathbb{R}^+} s^2\, ds \int_{|U|=1} e^{2\pi isP} \tilde{f}(sU)\, dU$$

Now, by the virtue of symmetry, we can rewrite the integral based on s as follows:

$$= \frac{1}{2} \int_{|U|=1} dU \int_{s\in\mathbb{R}} e^{2\pi isP} \tilde{f}(sU) s^2\, ds = \int_{|U|=1} F^{-1}\left\{\tilde{f}(sU)s^2\right\} dU$$

We don't have to perform convolution to simplify further and instead we use Fourier transform properties.

$$F^{-1}\left\{\tilde{f}(sU)\right\}(P) = \int_{S\in\mathbb{R}} e^{2\pi isP} \tilde{f}(sU)\, ds = \bar{f}(P,U)$$

Differentiate two times on both sides w.r.t P:

$$\frac{\partial \bar{f}(P,U)}{\partial P} = \int_{S\in\mathbb{R}} e^{2\pi isP} (2\pi is) \tilde{f}(sU)\, dS$$

$$\frac{\partial^2 \bar{f}(P,U)}{\partial P^2} = \int_{S\in\mathbb{R}} e^{2\pi isP} (-4\pi^2 s^2) \tilde{f}(sU)\, dS$$

Therefore, the value of $F^{-1}\left\{\tilde{f}(sU)s^2\right\}$ is given as follows:

$$F^{-1}\left\{\tilde{f}(sU)s^2\right\} = \frac{-1}{4\pi^2} \bar{f}_{PP}(P,U)$$

Therefore, our final result is given as:

$$f(x,y,z) = -\frac{1}{8\pi^2} \int_{|U|=1} \bar{f}_{PP}(P,U)\, dU$$

For the same three-dimensional case, if we assume the Fourier kernel to be $is(U^T X)$ instead, we land on quite a beautiful result. As per usual, keeping all other parameters the same:

$$f(x,y,z) = \int_{s\in\mathbb{R}^+} s^2\, ds \int_{|U|=1} e^{is(U^T X)} \tilde{f}(sU)\, dU$$

Setting $U^T X = P$:

$$f(x,y,z) = \int\limits_{s\in\mathbb{R}^+} s^2\, ds \int\limits_{|U|=1} e^{isP}\tilde{f}(sU)\, dU$$

By applying symmetry and swapping the integral signs, we observe the following:

$$f(x,y,z) = \frac{1}{2}\int\limits_{|U|=1} dU \int\limits_{s\in\mathbb{R}} e^{isP}\tilde{f}(sU)s^2\, ds = \frac{1}{2}\int\limits_{|U|=1} F^{-1}\{s^2\tilde{f}(sU)\}\, dU$$

Here, $F^{-1}\{s^2\tilde{f}(sU)\}$ can be calculated fairly quickly:

$$-\frac{\partial^2\bar{f}(P,U)}{\partial P^2} = -\overline{f_{PP}}(P,U) = \int\limits_{s\in\mathbb{R}} e^{isP}s^2\tilde{f}(sU)ds$$

Thus, the final result in this case is given as:

$$f(x,y,z) = -\frac{1}{2}\int\limits_{|U|=1} \left[\overline{f_{PP}}(P,U)\right]_{U\cdot X=P} dU$$

Let us now discuss a generalized case where we assume n to be an even number and $|X| = n$. To find the inverse Radon transform such that it results in $f(X)$. Let us firstly write down this scenario as a double integral as always:

$$f(X) = \int\limits_{s\in\mathbb{R}^+} s^{n-1}\, ds \int\limits_{|U|=1} e^{2i\pi s(U^T X)}\tilde{f}(sU)\, dU$$

Setting $U^T X = P$:

$$f(X) = \int\limits_{s\in\mathbb{R}^+} s^{n-1}\, ds \int\limits_{|U|=1} e^{2i\pi sP}\tilde{f}(sU)\, dU$$

By the virtue of symmetry, we can change the interval of s to $\mathbb{R}$ and also, we swap the integral signs:

$$f(X) = \frac{1}{2}\int\limits_{|U|=1} dU \int\limits_{s\in\mathbb{R}} e^{2i\pi sP}\tilde{f}(sU)s^{n-1}\, ds = \frac{1}{2}\int\limits_{|U|=1} F^{-1}\{s^{n-1}\tilde{f}(sU)\}\, dU$$

Using Inverse Fourier transform properties to produce the required value:

$$\bar{f}(P,U) = \int\limits_{s\in\mathbb{R}} e^{2i\pi sP}\tilde{f}(sU)\, ds$$

$$\frac{\partial^{n-1}\bar{f}(P,U)}{\partial P^{n-1}} = \int_{s\in\mathbb{R}} e^{2i\pi sP}(2i\pi s)^{n-1}\tilde{f}(sU)\,dU$$

$$F^{-l}\{s^{n-1}\tilde{f}(sU)\} = \frac{i^{n-1}}{(2\pi)^{n-1}}\frac{d^{n-1}\bar{f}(P,U)}{dP^{n-1}}$$

Therefore, our final result can be stated as such:

$$f(X) = \frac{i^{n-1}}{2(2\pi)^{n-1}}\int_{|U|=1} D_P^{(n-1)}\bar{f}(P,U)\,dU$$

The next logical step would be to generalize the inversion result for all odd $|X|$. Let, $|X| = n$ and n be some odd number.

$$f(X) = \int_{s\in\mathbb{R}^+} |s|^{n-1}\,ds \int_{|U|=1} e^{-2i\pi s(U^TX)}\tilde{f}(sU)\,dU$$

As always, setting $U^TX = P$ and making use of the symmetry about origin:

$$f(X) = \frac{1}{2}\int_{s\in\mathbb{R}} |s|^{n-1}\,ds \int_{|U|=1} e^{-2\pi isP}\tilde{f}(sU)\,dU$$

$$= \frac{1}{2}\int_{|U|=1} dU \int_{S\in\mathbb{R}} e^{-2\pi isP}|s|^{n-1}\tilde{f}(sU)\,ds = \frac{1}{2}\int_{|U|=1} dU\,F^{-1}\{\tilde{f}(sU)|s|^{n-1}\}$$

Here, the Fourier inverse can be calculated as such:

$$F^{-1}\{\tilde{f}(sU)|s|^{n-1}\} = F^{-1}\{\tilde{f}(sU)\} * F^{-1}\{|s|^{n-1}\}$$

$$= F^{-1}\{\tilde{f}(sU)\} * F^{-1}\{s^{n-1}\} * F^{-1}\{sgn\,s\}$$

The above expression can be simplified further:

$$F^{-1}\{\tilde{f}(sU)|s|^{n-1}\} = \bar{f}(P,U) * a_n\delta^{(n-1)}(P) * PV\left(\frac{1}{P}\right)$$

$$= a_n D_P^{(n-1)}\bar{f}(P,U) * PV\left(\frac{1}{P}\right)$$

Substituting the attained expression back into original inverse transform integral.

$$f(X) = \frac{a_n}{2} \int_{|U|=1} D_P^{(n-1)} \bar{f}(P,U) * PV\left(\frac{1}{P}\right) dU$$

$$f(X) = \frac{a_n}{2} \int_{|U|=1} dU \int_{T\in\mathbb{R}} \frac{D_T^{(n-1)} \bar{f}(T,U)}{P-T} dT$$

$$f(X) = -\frac{a_n}{2} \int_{|U|=1} dU \int_{T\in\mathbb{R}} \frac{D_T^{(n-1)} \bar{f}(T,U)}{T-P} dT$$

Here, we substitute $U^T X = U \cdot X = P$ back into the given double integral, which helps in rewriting the inverse relationship as a Hilbert transform:

$$f(x) = -\frac{a_n}{2} \int_{|U|=1} dU \int_{T\in\mathbb{R}} \frac{D_T^{(n-1)} \bar{f}(T,U)}{T - U \cdot X} dT$$

$$f(x) = -\frac{a_n}{2} \int_{|U|=1} dU\, H\left\{ D_P^{(n-1)} \bar{f}(P,U) \middle| U^T X = P \right\}$$

In this case, $a_n = \frac{-i^n}{(2\pi)^{n-1}}$. The normalization constant a_n will change w.r.t to the assumed Fourier kernel. So ultimately, our results can be summarized as follows:

$$f(X) = \begin{cases} A_n \int_{|U|=1} H\left\{ D_P^{(n-1)} \bar{f}(P,U) \middle| U^T X = P \right\} dU \ ; \ n \in E \\ A_n \int_{|U|=1} \left[D_P^{(n-1)} \bar{f}(P,U) \right]_{U^T X = P} dU \ ; \ n \in O \end{cases}$$

Here, A_n is the normalization term which changes w.r.t to the Fourier kernel. E, O stand for set of positive even and odd numbers respectively.

With this done, we can now discuss Inner products, Adjoint Radon transforms and its properties at length.

Inner product and adjoint Radon transforms

Firstly, let us define an inner product as between two functions $F(P,U), g(P,U)$ as follows (where parameters P, U have their usual meanings):

$$\langle F, g\rangle = \int\limits_{U\in S^{n-1}} \int\limits_{P\in\mathbb{R}} F(P,U)g(P,U)dP\,dU$$

Considering, $|U| = n$ and U is described on a $(n-1)$-sphere region of S^{n-1}. Now, let us assume that $R\{f(X)\}(P,U) = F(P,U)$ such that $|X| = n$. This assumption allows us to restate the above definition:

$$\langle Rf, g\rangle = \int\limits_{U\in S^{n-1}} \int\limits_{P\in\mathbb{R}} Rf(P,U)g(P,U)\,dP\,dU$$

By using the standard integral definition of Radon transform, let us extrapolate the above inner product as such:

$$\langle Rf, g\rangle = \int\limits_{U\in S^{n-1}} \int\limits_{P\in\mathbb{R}} \left[\int\limits_{X\in\mathbb{R}^n} f(X)\delta(P - U^T X)\,dX\right] g(P,U)\,dP\,dU$$

By invoking Fubini's theorem, we swap the integral signs:

$$\langle Rf, g\rangle = \int\limits_{U\in S^{n-1}} \int\limits_{X\in\mathbb{R}^n} \left[\int\limits_{P\in\mathbb{R}} g(P,U)\delta(P - U^T X)\,dP\right] f(X)\,dX\,dU$$

$$\langle Rf, g\rangle = \int\limits_{U\in S^{n-1}} \int\limits_{X\in\mathbb{R}^n} g(U^T X, U)f(X)\,dX\,dU$$

By swapping the integral signs once again, we obtain the following:

$$\langle Rf, g\rangle = \int\limits_{X\in\mathbb{R}^n} f(X)\left[\int\limits_{U\in S^{n-1}} g(U^T X, U)\,dU\right] dX$$

We define inner integral as the adjoint Radon transform denoted using R^*:

$$\langle Rf, g\rangle = \int\limits_{X\in\mathbb{R}^n} f(X)R^*g(X)\,dX = \langle f, R^*g\rangle$$

To be more formal, the end result is stated as such:

$$\langle Rf, g\rangle_{S^{n-1}\times\mathbb{R}} = \langle f, R^*g\rangle_{\mathbb{R}^n}$$

This is one of the most important results in the theory of Radon transforms and all of our upcoming discussion will be centred around it. Before proceeding further, let us quickly go through the definition of adjoint radon transform in two-dimensions.

$$R^*\{f(P,t)\}(x,y) = \int_0^\pi f(x\,cos\,t + y\,sin\,t\,, t)\,dt$$

Notice that for a given point (x, y) all possible lines are calculated in the given range of $t \in (0, \pi)$, this process is also called as smearing and because of this definition we intuitively realize that $R^*Rf \neq f$, which we will discuss later.

Now, let us discuss some crucial Inner-product identities related to adjoint Radon transforms.

(A) Fundamental identity : the identity that we had proved earlier.

$$\langle Rf, g\rangle_{S^{n-1}\times\mathbb{R}} = \langle f, R^*g\rangle_{\mathbb{R}^n}$$

(B) Symmetry identity: This identity becomes trivial once referring to the integral based definition presented earlier on.

$$\langle f, g\rangle_A = \langle g, f\rangle_A$$

(C) Polarization identity: This can be readily proved using fundamental identity and Symmetry identity.

$$\langle R^*Rf, g\rangle_{\mathbb{R}^n} = \langle g, R^*Rf\rangle_{\mathbb{R}^n} = \langle Rg, Rf\rangle_{S^{n-1}\times\mathbb{R}} = \langle Rf, Rg\rangle_{S^{n-1}\times\mathbb{R}}$$

Thus,

$$\langle R^*Rf, g\rangle_{\mathbb{R}^n} = \langle Rf, Rg\rangle_{S^{n-1}\times\mathbb{R}}$$

(D) Self-adjoint identity:

$$\begin{aligned}\langle R^*Rf, g\rangle_{\mathbb{R}^n} &= \langle g, R^*Rf\rangle_{\mathbb{R}^n} = \langle Rg, Rf\rangle_{S^{n-1}\times\mathbb{R}} = \langle Rf, Rg\rangle_{S^{n-1}\times\mathbb{R}} \\ &= \langle f, R^*Rg\rangle_{\mathbb{R}^n}\end{aligned}$$

Thus,

$$\langle R^*Rf, g\rangle_{\mathbb{R}^n} = \langle f, R^*Rg\rangle_{\mathbb{R}^n}$$

(E) Norm-based identities:

$$\langle R^*Rf, f\rangle_{\mathbb{R}^n} = \langle Rf, Rf\rangle_{S^{n-1}\times\mathbb{R}} = \int\limits_{P\in\mathbb{R}} \int\limits_{U\in S^{n-1}} \big(Rf(P,U)\big)^2 \, dU \, dP = \|Rf\|^2$$

$$\langle RR^*g, g\rangle_{S^{n-1}\times\mathbb{R}} = \langle R^*g, R^*g\rangle_{\mathbb{R}^n} = \int\limits_{X\in\mathbb{R}^n} \big(R^*g(X)\big)^2 \, dX = \|R^*g\|^2$$

(F) Positivity: These identities directly follow from the Norm-based identities.

$$\langle R^*Rf, f\rangle_{\mathbb{R}^n} = \|Rf\|^2 \geq 0$$

$$\langle RR^*g, g\rangle_{S^{n-1}\times\mathbb{R}} = \|R^*g\|^2 \geq 0$$

(G) Operator Norm-based result: Standard Hilbert space identities hold for any suitable $f \in I$ such that, I, J are Hilbert spaces with $R: I \to J$.

$$\|R\|^2 = \|R^*R\| = \|RR^*\|$$

We will now move forward with discussing Adjoint Radon transform convolution. For this, let us firstly write down the integral form of $R^*Rf(X)$ such that $|X| = 2$.

$$R^*Rf(X) = \int\limits_0^\pi Rf(U_t^T X, t) \, dt = \int\limits_0^\pi \int_{\mathbb{R}^2} f(y)\delta(U_t^T X - U_t^T y) \, dy \, dt$$

By using Fubini's theorem, we swap the integral signs and rewrite the integral-based expression.

$$R^*Rf(X) = \int\limits_{\mathbb{R}^2} f(y) \int\limits_0^\pi \delta\big(U_t^T (X - y)\big) \, dt \, dy$$

Rewrite the inner integral by assuming it is equivalent to a function $K(X - y)$.

$$R^*Rf(X) = \int\limits_{\mathbb{R}^2} f(y)K(X - y) \, dy$$

We set $w = X - y$, and focus solely on simplifying $K(w)$. After which, we will assume $w = (r\,cos\,a\,, r\,sin\,a)$ and it directly follows that $\|w\| = r$. (Recall that in this case $U_t = (cos\,t\,, sin\,t)$.)

$$K(w) = \int_0^\pi \delta(U_t^T w)\,dt = \int_0^\pi \delta(r\,cos\,t\,cos\,a - r\,sin\,t\,sin\,a)\,dt$$

$$K(w) = \int_0^\pi \delta(r\,cos(t-a))\,dt = \frac{1}{r}\int_0^\pi \delta(cos(t-a))\,dt$$

This is quite a well-known integral and it simply evaluates to the following:

$$K(w) = \frac{1}{r} = \frac{1}{\|w\|} = \frac{1}{\|X-y\|}$$

Which means $R^*Rf(X)$ now looks like the following:

$$R^*Rf(X) = \int_{y\in\mathbb{R}^2} \frac{f(y)}{\|X-y\|}\,dy = f(X) * \frac{1}{\|X\|}$$

Remember when we intuitively described that $R^*Rf(X) \neq f(X)$? above is the rigorous proof of why it is as such. In the case of $|X| = n$, the required formulation changes as follows:

$$R^*Rf(X) = \int_{y\in\mathbb{R}^n} \frac{f(y)}{\|X-y\|^{n-1}}\,dy = f(X) * \frac{1}{\|X\|^{n-1}}$$

Also, there is a nice Fourier interpretation for the n-dimensional case and is usually described as such:

$$F\{R^*Rf\}(E) = F\left\{f(X) * \frac{1}{\|X\|^{n-1}}\right\}(E) = F\{f(X)\}(E)F\left\{\frac{1}{\|X\|^{n-1}}\right\}(E)$$

By Riesz's potential formula, we know that $F\left\{\frac{1}{\|X\|^{n-1}}\right\}(E) = \frac{c_n}{\|E\|}$ such that c_n is normalization constant obtained after applying certain Fourier kernel.

$$F\{R^*Rf\}(E) = c_n\tilde{f}(E)\|E\|^{-1}$$

Here, $F\{R^*Rf\}(E)$ is generally given as $R^*\widetilde{Rf(E)}$. Ultimately, the result is given as:

$$R^*\widetilde{Rf(E)} = c_n\tilde{f}(E)\|E\|^{-1}$$

This result can be further extended to inner-products once again:

$$\langle R^*Rf, f\rangle_{\mathbb{R}^n} = \langle F(R^*Rf), F(f)\rangle_{\mathbb{R}^n} = c_n \int\limits_{E\in\mathbb{R}^n} \|E\|^{-1}\tilde{f}^2(E)dE$$

This can be verified very easily by observing that the positivity identity remains fulfilled.

$$\langle R^*Rf, f\rangle_{\mathbb{R}^n} = c_n \int\limits_{E\in\mathbb{R}^n} \|E\|^{-1}\left|\tilde{f}(E)\right|^2 dE \geq 0$$

Similarly, a nice inner product identity also follows:

$$\left\langle R^*\widetilde{Rf(E)}, \tilde{f}(E)\right\rangle_{\mathbb{R}^n} = \left\langle c_n\tilde{f}(E)\|E\|^{-1}, \tilde{f}(E)\right\rangle_{\mathbb{R}^n}$$

Now, we will be looking at various basic yet important properties of adjoint radon transforms.

(a) Linearity: Directly follows from the integral definition.

$$R^*\{F+g\}(X) = R^*\{F\}(X) + R^*\{g\}(X)$$

(b) Translation identity: adjoint Radon transforms scale as expected. (Such that $dim(A) = n \times 1$):

$$R\{f(X+A)\}(P,U) = F(P+U^TA,U)$$

$$R^*\{F(P+U^TA,U)\}(X) = \int\limits_{U\in S^{n-1}} F(U^TX+U^TA,U)\,dU$$

$$R^*\{F(P+U^TA,U)\}(X) = R^*F(X+A)$$

(c) Scaling identity: A natural scaling identity follows. (Such that $c \neq 0$.)

$$R^*\left\{F\left(\frac{P}{c},U\right)\right\}(X) = \int\limits_{U\in S^{n-1}} F\left(\frac{U^TX}{c},u\right)du = R^*F\left(\frac{X}{c}\right)$$

(d) Convolution structure: As mentioned earlier, R^*R can be expressed as a class of Riesz's potential.

$$\widetilde{R^*R}f(E) = c_n \frac{\tilde{f}(E)}{\|E\|} = c_n I_n$$

Where I_n is the Riesz's potential of nth order.

(e) Wavefront interpolation: Wavefronts assume that the bidirectional noises are to be neglected due to which the following result is observed. That is, the origin-based noise is ignored entirely.

$$WF\big(R^*Rf(X)\big) = WF\big(f(X) * \|X\|^{-(n-1)}\big) = WF\big(f(X)\big)$$

(f) Sobolev mapping: By using Sobolev properties, the following mapping can be constructed.

$$\|f\|_{H^s}^2 = \int_{E\in\mathbb{R}^n} (1+|E|^2)^s \big(F\{f(X)\}(E)\big)^2 \, dE = \int_{E\in\mathbb{R}^n} (1+|E|^2)^s \big|\tilde{f}(E)\big|^2 \, dE$$

Similarly, for R^*Rf we have the following:

$$\|R^*Rf\|_{H^{s+1}}^2 = \int_{E\in\mathbb{R}^n} (1+|E|^2)^s \big(F\{R^*Rf\}(E)\big)^2 \, dE$$

$$= \int_{E\in\mathbb{R}^n} (1+|E|^2)^{s+1} \left(\frac{1}{E}\big|\tilde{f}(E)\big|\right)^2 dE$$

For sufficiently large $|E|$, we have:

$$\|R^*Rf\|_{H^s}^2 \leq \int_{E\in\mathbb{R}^n} (1+|E|^2)^s |f(E)|^2 \, ds$$

Thus, it can be said that:

$$\|R^*Rf\|_{H^{s+1}}^2 \leq \|f\|_{H^s}^2$$

Which ultimately allows us to assert that:

$$R^*Rf\colon H^s(\mathbb{R}^n) \to H^{s+1}(\mathbb{R}^n)$$

(g) Imperfect inversion: As already discussed, R^*R results in given function and added noise. This is one of the most important properties of adjoint Radon transforms.

$$R^*Rf(X) = f(X) * \frac{1}{\|X\|^{n-1}} \neq f(X)$$

Even though R^*R leads to imperfect inversions, we can change that by introducing a special operator K. For the upcoming exposition, let us assume $|X| \in E$ and as a virtue, $KF(P,U) = A_n H\left(D_P^{n-1}F(P,U)\right)$, such that A_n is normalization constant described in accordance with the Fourier kernel and H is the standard Hilbert transform. Do bear in mind that $Rf(P,U) = F(P,U)$.

$$R^*\big(KF(P,U)\big)(X) = R^*\left(A_n H\big(D_P^{n-1}F(P,U)\big)\right)(X)$$

Now, writing the above expression by using the integral identity of R^*:

$$R^*\big(KF(P,U)\big)(X) = \int_{U \in S^{n-1}} A_n H\left(D_p^{n-1}F(P,U)|P \to U^T X\right) dU$$

Now, realize that the above expression is basically the integral definition of R^{-1} when $|X| \in E$. This allows us to rewrite the above as such:

$$R^*\big(KF(P,U)\big)(X) = R^{-1}\{F(P,U)\}(X) = f(X)$$

Similarly, for $|X| \in O$, let us assume $KF(P,U) = A_n D_P^{n-1}\{F(P,U)\}$ and calculate the value of R^*K:

$$R^*\big(KF(P,U)\big)(X) = R^*\big(A_n D_P^{n-1}F(P,U)\big)(X)$$

Once again, on using the integral identity of R^*, we have:

$$R^*\big(KF(P,U)\big)(X) = A_n \int_{U \in S^{n-1}} [D_P^{n-1}F(P,U)]_{P=U^T X}\, dU$$

Recognize that the above structure is equivalent to R^{-1}, because of which we have:

$$R^*\big(KF(P,U)\big)(X) = R^{-1}\{F(P,U)\}(X) = f(X)$$

Thus, to summarize these findings, we can firstly define the K operator as follows:

$$KF(P,U) = \begin{cases} A_n H\left(D_P^{n-1} F(P,U)\right) ;\ n \in E \\ A_n D_P^{n-1} F(P,U) ;\ n \in O \end{cases}$$

To describe the workings of this operator, let us assume $KF(P,U) = \dot{F}(P,U)$ and $Rf(P,U) = F(P,U)$.

$$R^*\big(KF(P,U)\big)(X) = R^*\left(\dot{F}(P,U)\right)(X) = \int_{U \in S^{n-1}} \dot{F}(U^T X, U)\, dU$$

$$= R^{-1}\{F(P,U)\}(X) = f(X)$$

As per many older texts, K is often referred to as the filter operator as it helps in removing noise and nullifying the $\|X\|^{-(n-1)}$ noise term. In short, the role of K operator can be described as follows:

$$R^* KF = R^{-1} F$$

Also, a lot of texts like to introduce the term backprojection denoted with B. To aptly put, backprojection and adjoint Radon transforms are identical terms used quite interchangeably in many of the modern texts. To end this section, let us quickly state few backprojection formulae and relevant identities.

$$f(X) = B\big(F(P,U)\big)(X) = \int_{U \in S^{n-1}} F(U^T X, U)\, dU$$

In the case of $|X| = 2$, we have the known standard identity:

$$f(x,y) = B\big(F(P,t)\big)(x,y) = \int_{t=0}^{\pi} F(x \cos t + y \sin t\,, t)\, dt$$

Ultimately, the following can be stated to be true up to normalization constants:

$$R^*\{F(P,U)\}(X) \equiv\ B\big(F(P,U)\big)(X)$$

<u>Example 6.1:</u> Compute $R^* R\{\delta(X - A)\}$ such that $|X|, |A| = n$ and the transform parameters for R are (P,U).

Firstly, let us compute $R\{\delta(X - A)\}(P,U)$:

$$R\{\delta(X - A)\}(P,U) = \int_{X \in \mathbb{R}^n} \delta(X - A)\delta(P - U^T X)\, dX = \delta(P - U^T A)$$

Now, we can compute $R^*\{\delta(P - U^T A)\}(X)$:

$$R^*\{\delta(P - U^T A)\}(X) = \int_{U \in S^{n-1}} \delta(U^T X - U^T A)\, dU = \int_{U \in S^{n-1}} \delta\big(U^T (X - A)\big)\, dU$$

$$R^*\{\delta(P - U^T A)\}(X) = \frac{1}{\|X - A\|^{n-1}}$$

Thus, the final solution is stated as such:

$$R^* R\{\delta(X - A)\} = \frac{1}{\|X - A\|^{n-1}}$$

This completes our theoretical discussion of Radon transforms, in the next section, we will be discussing the applications of Radon transforms and its usage in solving mathematical problems.

Applications of Radon transforms

The most important application of Radon transforms is in solving the heat equation in n-dimensions as it helps in transforming the n-dimensional problem into a one-dimensional problem. The heat PDE in n-dimensions is stated as follows:

$$u_t = c\Delta u$$

Such that, $\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \cdots + \frac{\partial^2 u}{\partial x_n^2}$ and $c > 0$.

Assume, $X = [x_i], V = [v_i]\ \forall i \le n$. Using which the Radon transform of function u is defined as such.

$$R\{u(X,t)\}(P,V) = \int_{X \in \mathbb{R}^n} u(X,t)\delta(P - V^T X)\, dX = \hat{u}(P,V;t)$$

The Radon transform of u_t can be calculated with ease as follows:

$$R\{u_t(X,t)\}(P,V) = \frac{\partial}{\partial t} R\{u(X,t)\}(P,V) = \hat{u}_t$$

The following property has been studied quite extensively in the section of *computations in Radon transforms*:

$$R\{\Delta u\}(P,V) = R\left\{\sum_{i\leq n} u_{x_i x_i}\right\}(P,V) = \|V\|\hat{u}_{PP} = \hat{u}_{PP}$$

Thus, applying Radon transform to the original heat equation will result in the following expression:

$$\hat{u}_t = c\hat{u}_{PP}$$

We will now be defining Fourier transform of the function $\hat{u}$.

$$F\{\hat{u}(P;t)\}(k) = \frac{1}{2\pi}\int_{P\in\mathbb{R}} e^{-i} \quad \hat{u}(P;t)\,dP = \tilde{\hat{u}}(k;t)$$

Consequently, the Fourier transform of $\hat{u}_t$ and $\hat{u}_{PP}$ are also calculated.

$$F\{\hat{u}_t(P;t)\}(k) = \frac{\partial}{\partial t}F\{\hat{u}(P;t)\}(k) = \tilde{\hat{u}}_t(k;t)$$

$$F\{\hat{u}_{PP}(P;t)\}(k) = (ik)^2 F\{\hat{u}(P;t)\}(k) = -k^2\tilde{\hat{u}}(k;t)$$

Applying Fourier transform to the derived one-dimensional heat equation results in a first order ODE w.r.t the time parameter.

$$\tilde{\hat{u}}_t + ck^2\tilde{\hat{u}} = 0$$

Multiplying the exponential term e^{ck^2t} on both sides of the equation and simplifying further results in the following:

$$e^{ck^2t}\tilde{\hat{u}}_t + e^{ck^2t}\left(ck^2\tilde{\hat{u}}\right) = 0$$

$$\frac{d}{dt}\left(e^{ck^2t}\tilde{\hat{u}}\right) = 0 \Rightarrow \tilde{\hat{u}} = e^{-ck^2t}Q(k)$$

If we assume the initial condition to be $u(X,0) = f(X)$. Then its also fair to imply, $\hat{u}(P;0) = \hat{f}(P)$ and $\tilde{\hat{u}}(k;0) = \tilde{\hat{f}}(k)$. Because of which we can say, $Q(k) = \tilde{\hat{f}}(k)$.

$$\tilde{\hat{u}} = e^{-ck^2t}\tilde{\hat{f}}(k)$$

Now, we will be taking Fourier inverse on both sides and making use of the Fourier convolution.

$$\hat{u} = F^{-1}\left\{e^{-ck^2t}\tilde{\hat{f}}(k)\right\}(P,V) = F^{-1}\left\{e^{-ck^2t}\right\}(P,V) * F^{-1}\left\{\tilde{\hat{f}}(k)\right\}(P,V)$$

The Fourier inverse of e^{-ck^2t} is a well-known one, given as:

$$F^{-1}\left\{e^{-ck^2t}\right\}(P) = \frac{1}{2\sqrt{\pi ct}}e^{-\frac{P^2}{4ct}} = G(P,t)$$

Thus, the inverse Fourier-based expression can be simplified as such:

$$\hat{u} = G(P,t) * \hat{f}(P)$$

Now, we take Inverse Radon transform on both sides:

$$u = R^{-1}\left\{G(P,t) * \hat{f}(P)\right\}(X) = R^{-1}\{G(P,t)\}(X) * R^{-1}\left\{\hat{f}(P)\right\}(X)$$

Here, the Radon inverse of the generalized heat kernel is given as follows:

$$R^{-1}\left\{\frac{1}{\sqrt{4\pi ct}}e^{-\frac{P^2}{4ct}}\right\}(X) = \frac{1}{(4\pi ct)^{n/2}}e^{-\frac{\|X\|^2}{4ct}} = G^*(X,t)$$

Therefore, our final result is stated as such:

$$u(X,t) = \left(\frac{1}{(4\pi ct)^{n/2}}e^{-\frac{\|X\|^2}{4ct}}\right) * f(X) = \frac{1}{(4\pi ct)^{n/2}}\int\limits_{T\in\mathbb{R}} e^{-\frac{\|T\|^2}{4ct}}f(X-T)\,dT$$

Or alternatively, the same can also be written in a compact form:

$$u(X,t) = G^*(X,t) * f(X) = \int\limits_{T\in\mathbb{R}} G^*(T,t)f(X-T)\,dT$$

Radon transform is essentially a dimensionality reducer tool when it comes to solving PDEs and finding general solutions. Similarly, we will also be attempting to solve the wave equation PDE in nD by setting the initial conditions to be $u(X,0) = f(X)$ and $u_t(X,0) = g(X)$.

$$u_{tt}(X,t) = c^2 \Delta u(X,t)$$

Such that, c is some real quantity. Assuming $R\{u(X,t)\}(P,U) = v(P,U,t)$, we take Radon transform on both sides:

$$v_{tt}(P,U,t) = c^2 v_{PP}(P,U,t)$$

The general solution to this PDE is generally given as follows by assuming that $R\{u(X,0)\}(P,U) = R\{f(X)\}(P,U) = F(P,U)$ and $R\{u_t(X,0)\}(P,U) = R\{g(X)\}(P,U) = G(P,U)$:

$$v(P,t) = \frac{1}{2}[F(P+ct,U) + F(P-ct,U)] + \frac{1}{2c}\int_{P-ct}^{P+ct} G(y,U)\,dy$$

The final solution for this can be simply stated as:

$$= \frac{1}{2}R^{-1}\{F(P+ct,U) + F(P-ct,U)\}(X) + \frac{1}{2c}R^{-1}\left\{\int_{P-ct}^{P+ct} G(y,U)\,dy\right\}(X)$$

And in case of a completely general solution where initial values are not assigned, the solution to a nD wave equation is given as follows (Assume, M, N to be appropriately well-defined functions)

$$v(P,U,t) = M(P+ct,U) + N(P-ct,U)$$

$$u(X,t) = R^{-1}\{M(P+ct,U)\}(X) + R^{-1}\{N(P-ct,U)\}(X)$$

Other than PDEs, Radon transform is also quite helpful in solving a particular kind of integral equations also called as eigenvalue problems. For the given integral equation, our goal is to find h in the given function space of $L^2(\mathbb{R}^n)$.

$$\int_{T\in\mathbb{R}} \frac{h(T)}{\|X-T\|^{n-1}}\,dx = h(X)$$

By referring to our previous derivations, it becomes easy to state the following:

$$R^*Rh(X) = \int_{T\in\mathbb{R}} \frac{h(T)}{\|X-T\|^{n-1}}\,dx$$

Thus, the original expression turns into the following:

$$R^*Rh(X) = h(X)$$

Complying to our previous expositions, we apply the Fourier transform on both sides with respect to E to obtain the following standard result:

$$c_n\|E\|^{-1}\hat{h}(E) = \hat{h}(E)$$

$$\hat{h}(E)(c_n\|E\|^{-1} - 1) = 0$$

For the above equation to satisfy, only two possible ways exist. Either $\hat{h}(E) = 0$ or $c_n\|E\|^{-1} - 1 = 0$. By extrapolating the second route, we realize the following to hold true:

$$c_n\|E\|^{-1} - 1 = 0 \;\Rightarrow\; \|E\| = c_n$$

Meaning the support of $\hat{h}(E)$ must lie on the boundary of this n-sphere with radius c_n.

$$supp\left(\hat{h}(E)\right) \subseteq \{E : \|E\| = c_n\}$$

But this is a distribution and cannot be represented by any of the standard functions within the function space of $L^2(\mathbb{R}^n)$. Therefore, the only true solution to this problem is the trivial solution of $\hat{h}(E) = 0$ which directly implies $h(X) = 0$.

Example 7.1: For the given eigenvalue problem $R^*Rh = h$, find a suitable distribution of $\hat{h}(E)$ that satisfies the given equation.

By referring to the above exposition, we rewrite an important statement.

$$supp\left(\hat{h}(E)\right) \subseteq \{E : \|E\| = c_n\}$$

By using the δ function, we can write a suitable distribution as follows:

$$\hat{h}(E) = \delta(\|E\| - c_n)$$

It can be seen that this distribution satisfies the constraint $\hat{h}(E)(c_n\|E\|^{-1} - 1) = 0$ and results as nil when $\|E\| = c_n$. Thus, this is an appropriate distribution. To be slightly more sophisticated, distributions of the following form are acceptable as long as l is finite and non-zero. (Note: this is not the generalized version of the solution space.)

$$\hat{h}(E) = l(E)\delta(\|E\| - c_n)$$

Moving on forward, let us discuss one last important aspect within applications of Radon transforms.

Based on the given Radon transform relationship, we can also predict the conditions imposed on the transformed function. Let us say, we are told that

$F_{PP}(P,U) = 0$ and we also know that, $R\{f(X)\}(P,U) = F(P,U)$, we are required to find out the conditions imposed on the given transformed function f.

As we already know $F_{PP} = 0$, we can extrapolate this as follows:

$$F_{PP}(P,U) = \|U\| F_{PP}(P,U) = R\{\Delta f(x)\}(P,U) = 0$$

Taking inverse Radon transform in the above expression ultimately leads to the following:

$$\Delta f(X) = 0$$

This is essentially the dual problem of solving PDEs. Here, we are given the functional equation and we are supposed to find the corresponding PDE. Next up, let us consider the functional equation to be, $aF_{PP}(P,U) + bF_{PP}(-P,-U) = 0$ such that $a,b > 0$ and $R\{f(X)\}(P,U) = F(P,U)$. Firstly, the following is a crucial identity:

$$F(P,U) = \int_{U^T X = P} f(X)\, dX = \int_{(-U)^T X = -P} f(X)\, dX = F(-P,-U)$$

Using this identity, we can rewrite our original equation as follows:

$$(a+b)F_{PP}(P,U) = 0$$

Based on the given information, we can state the following:

$$(a+b)F_{PP}(P,U) = (a+b)\|U\|^2 F_{PP}(P,U) = (a+b)R\{\Delta f(X)\}(P,U) = 0$$

Taking inverse Radon transform on both sides gives us the final result.

$$(a+b)\Delta f(X) = 0$$

That wraps up the mathematical applications of Radon transforms. Up next, we will be discussing about exponential Radon transforms and the relevant derivations and properties.

Exponential Radon transforms

An exponential Radon transform of $f(X)$ such that $|X| = n$ and $n \geq 2$ can be defined as follows:

$$R_l f(P,U) = \int_{X \in \mathbb{R}^n} e^{l(U^T X)} f(X) \delta(P - U^T X)\, dX$$

Assuming the transform is being taken on the hyperplane $U^T X = P$, The above expression can be simplified as follows:

$$R_l f(P,U) = e^{lP} \int\limits_{X \in \mathbb{R}^n} f(X)\delta(P - U^T X)\, dX = e^{lP} Rf(P,U)$$

The relationship between exponential radon transform and standard Radon transform often referred to as the conjugation property, is one of the most important ones in this entire segment.

$$R_l f(P,U) = e^{lP} Rf(P,U) = R\left\{e^{l(U^T X)} f(X)\right\}(P,U)$$

The above can also be rewritten using operator theory. Let, the operator M_l possess the property of $M_l g = e^{lP} g$, where g is some suitable function:

$$R_l f(P,U) = M_l Rf(P,U)$$

Let us now discuss few important properties of exponential Radon transforms:

(A) Zero order exponential Radon transform:

$$R_0 f(P,U) = \int\limits_{X \in \mathbb{R}^n} f(X)\delta(P - U^T X)\, dX = Rf(P,U)$$

(B) Direct relationship with Fourier transform: Assume exponential Radon transform of order $-il$.

$$R_{-il} f(P,U) = \int\limits_{X \in \mathbb{R}^n} e^{-ilU^T X} f(X)\delta(P - U^T X)\, dX = F\{f(X)\delta(P - U^T X)\}(lU)$$

(C) Orthogonality: Assume, $O = [0]\ \forall i \leq n$.

$$R_l\{\delta(x)\}(P,U) = \int\limits_{X \in \mathbb{R}^n} e^{lU^T X} \delta(X)\delta(P - U^T X)\, dX = e^{lU^T X}\delta(P - U^T X)|_{X=O}$$

$$R_l\{\delta(x)\}(P,u) = \delta(P)$$

(D) Orthogonality invariance property: The above expression can also be rewritten as follows:

$$R_l\{\delta(x)\}(P,U) = R\{\delta(x)\}(P,U)$$

(E) Differentiation under the transform sign:

$$D_P R_l f(P,U) = \int\limits_{X\in\mathbb{R}^n} f(X) D_P\left(e^{lP}\delta(P - U^T X)\right) dX$$

$$= l \int\limits_{X\in\mathbb{R}^n} e^{lP} f(X)\delta(P - U^T X)\, dX - \int\limits_{X\in\mathbb{R}^n} e^{lP} f(X) D_P\big(\delta(P - U^T X)\big)\, dX$$

Which can be compressed to present the following identity:

$$D_P R_l f(P,U) = l R_l f(P,U) - e^{lP} D_P\big(Rf(P,U)\big)$$

With the properties done, let us now discuss the Fourier slice theorem, presented using the exponential Radon transform. Firstly, recall the Fourier slice theorem given in standard Radon transforms.

$$F_y\{Rh(y,U)\}(s) = F\{h(X)\}(sU)$$

Here, let us assume $h(X) = e^{l(U^T X)} f(X)$. Because of which, $R_l f(P,U) = Rh(P,U)$. This assumption, leads to the following equations:

$$F_y\{Rh(y,U)\}(s) = F_y\{e^{lP} Rf(y,U)\}(s) = F_y\{R_l f(y,U)\}(s)$$

$$F\{h(X)\}(sU) = F\left\{e^{l(U^T X)} f(X)\right\}(sU)$$

The second equation can be simplified further by using Fourier transform properties:

$$F\left\{e^{l(U^T X)} f(X)\right\}(sU) = \int\limits_{X\in\mathbb{R}^n} e^{-isU^T X} e^{l(U^T X)} f(X)\, dX$$

$$= \int\limits_{X\in\mathbb{R}^n} e^{-i(s+il)U^T X} f(X)\, dX = F\{f(X)\}\big((s+il)U\big)$$

Thus, the revised Fourier slice theorem w.r.t exponential Radon transforms, is asserted as such:

$$F_y\{R_l f(y,U)\}(s) = F\{f(X)\}\big((s+il)U\big)$$

Similarly, the adjoint exponential Radon transform can be derived quite easily by using the known inner product result. Let, $h(X) = e^{l(U^T X)} f(X)$ as per usual.

$$\langle Rh, e^{lP} v\rangle_{S^{n-1}\times\mathbb{R}} = \langle h, R^*(e^{lP} v)\rangle_{\mathbb{R}^n}$$

As we already now, $Rh(P,U) = R_l f(P,U)$.

$$\langle R_l f, e^{lP} v\rangle_{S^{n-1}\times\mathbb{R}} = \langle h, R^*(e^{lP} v)\rangle_{\mathbb{R}^n}$$

Here, we specify that $R^*(e^{lP} v) = R_l^*(v)$. Because of which, we get the following important inner product identity:

$$\langle R_l f, e^{lP} v\rangle_{S^{n-1}\times\mathbb{R}} = \left\langle e^{l(U^T X)} f, R_l^* v\right\rangle_{\mathbb{R}^n}$$

Therefore, adjoint exponential Radon transform can formally be defined as follows:

$$R_l^* F(P, U) = \int\limits_{U\in S^{n-1}} e^{lU^T X} F(U^T X, U)\, dU$$

Similarly, a mixed convolution trick in exponential Radon transforms is given as such. Let, $h(X) = e^{l(U^T X)} f(X)$.

$$R^* R_l f(X) = R^*(e^{lP} Rf)(X) = R^* Rh(X)$$

$$= h(X) * \frac{1}{\|X\|^{n-1}} = e^{l(U^T X)} f(X) * \frac{1}{\|X\|^{n-1}}$$

Example 8.1: Assume that $h(X) = e^{l(U^T X)} f(X)$ and $Rf(P, U) = F(P, U)$. Compute the value of $R_l^* R\left(e^{-l(U^T X)} f(X)\right)$.

Firstly, computing the value of the inner Radon transform:

$$R\left(e^{-l(U^T X)} f(X)\right) = \int\limits_{X\in\mathbb{R}^n} e^{-l(U^T X)} f(X)\delta(P - U^T X)\, dx = e^{-lP} F(P, U)$$

Now simplifying rest of the expression by using the integral definition of adjoint exponential Radon transform.

$$R_l^t\left(e^{-lP} F(P, U)\right) = \int\limits_{U\in S^{n-1}} F(U^T X, U)\, dU = R^* Rf(X) = f(X) * \frac{1}{\|X\|^{n-1}}$$

Thus, our final answer can be stated as such:

$$R_l^* R\left(e^{-l(U^T X)} f(X)\right) = f(X) * \frac{1}{\|X\|^{n-1}}$$

What ultimately remains for discussion in this segment is the inverse of exponential Radon transform. To procedurally compute this, as always assume that $|X| = n$, $h(X) = e^{l(U^T X)} f(X)$, $Rh(P, U) = H(P, U)$ and $Rf(P, U) = F(P, U)$. We already know that:

$$Rh(P, U) = R\left(e^{l(U^T X)} f(X)\right)(P, U) = e^{lP} Rf(P, U) = e^{lP} F(P, U)$$

Now, taking inverse Radon transform on both sides:

$$R^{-1} H(X) = R^{-1}\{e^{lP} F(P, U)\}(X)$$

Here, we specify that $R^{-1}\{e^{lP} F(P, U)\}(X) = R_l^{-1}\{F(P, U)\}(X)$ because of which we have:

$$h(X) = R_l^{-1}\{F(P, U)\}(X)$$

Referring back to the inverse Radon transform definition, we can state the inverse exponential Radon transform with ease.

$$R_l^{-1}\{F(P, U)\}(X) = \begin{cases} A_n \displaystyle\int_{U \in S^{n-1}} H\left(D_P^{n-1}\left(e^{lP} F(P, U)\right) | P \to U^T X\right) dU\ ;\ n \in E \\ A_n \displaystyle\int_{U \in S^{n-1}} \left[D_P^{n-1}\left(e^{lP} F(P, U)\right)\right]_{U^T X = P} dU\ ;\ n \in O \end{cases}$$

Let us solve one more example before going to the next section.

<u>Example 8.2:</u> Given that $h(X) = e^{l(U^T X)} f(X)$, $|X| = n$ and $Rh(P, u) = H(P, U)$. Prove or disprove that, $R_l f(P, U) = H(P, U) = H(-P, -U)$.

Let us make use of the integral definition of the exponential Radon transform.

$$R_l f(P, U) = \int_{X \in \mathbb{R}^n} e^{l(U^T X)} f(X) \delta(P - U^T X)\, dX = \int_{U^T X = P} e^{l(U^T X)} f(X)\, dX$$

By using the already given information, we have:

$$R_l f(P, U) = \int_{U^T X = P} h(X)\, dX = \int_{(-U)^T X = -P} h(X)\, dX = H(-P, -U)$$

Thus, we assert the following result:

$$R_l f(P, U) = H(P, U) = H(-P, -U)$$

This completes our discussion of continuous Radon transforms; In the next section we will discuss ways in which Discrete Radon transforms can be constructed and their relevant properties.

Discrete Radon transforms

There are multiple definitions of a discrete Radon transform and we will try to study as many as we can. To study our first variation, we will be deriving it from the definition of Radon transform itself. Let us start with the following setup:

$$R\{\delta(x-\lfloor x\rfloor, y-\lfloor y\rfloor)f(x,y)\}(P,U) = \int\limits_{\substack{x,y\in R^2 \\ U^TX=P}} \delta(x-\lfloor x\rfloor, y-\lfloor y\rfloor)f(x,y)dx\,dy$$

As we have already studied the properties of delta function in detail, we can state the following simplification with ease:

$$= f(x,y)\delta(P-U^TX)|_{x,y\in\mathbb{Z}^2} = \sum_{x,y\in\mathbb{Z}^2} f(x,y)\delta(P-U^TX)$$

As we are discussing a two-dimensional case, we can simply write $U^TX = x\,cos\,t + y\,sin\,t$ and define the following expression as a discrete Radon transform.

$$R_Df(P,t) = \sum_{x,y\in\mathbb{Z}^2} f(x,y)\delta(P-x\,cos\,t-y\,sin\,t)$$

If the floor-based delta function notation looks confusing, then the following notation can be followed alternatively:

$$\delta(x-\lfloor x\rfloor, y-\lfloor y\rfloor) = \sum_{m,n\in\mathbb{Z}^2} \delta(x-m, y-n)$$

So ultimately, the bridge between Radon transform and discrete Radon transform can be stated as follows:

$$R\{\delta(x-\lfloor x\rfloor, y-\lfloor y\rfloor)f(x,y)\}(P,t) = R_Df(P,t)$$

An alternative method to define the discrete Radon transform would be by considering the whole transform as a series itself.

$$R_D f(P,t) = \sum_{\substack{x,y \in \mathbb{Z}^2 \\ x\cos t + y \sin t}} f(x,y)$$

Let us define an indication function $M(x,y)$ to map the constraints of the given sum properly.

$$M(x,y) = \begin{cases} 1\,;\; x\cos t + y\sin t = P \\ 0\,;\; x\cos t + y\sin t \neq P \end{cases}$$

Using this indicator function M, we can simplify the original expression as follows:

$$R_D f(P,t) = \sum_{x,y \in \mathbb{Z}^2} f(x,y)M(x,y) = \sum_{x,y \in \mathbb{Z}^2} f(x,y)\delta(P - x\cos t - y\sin t)$$

This definition becomes quite practical as we can freely define the indicator function M while relaxing the line-based constraints. For example, consider ε to be some threshold limit, based on this an indicator function is constructed as follows:

$$M_\varepsilon(x,y) = \begin{cases} 1\,;\, |x\cos t + y\sin t - P| \leq \varepsilon \\ 0\,;\, |x\cos t + y\sin t - P| > \varepsilon \end{cases}$$

Now to define another class of discrete Radon transforms, we will assume the line constraint to be $y = tx + P$. Also, the function that is being transformed is periodic in nature for given integer parameters λ, μ and Ω be some prime and x, y are defined on the discrete domain of 0 to $\Omega - 1$.

$$f(x,y) = f(x + \lambda\Omega, y + \mu\Omega)$$

The whole point of this discretizing scheme is to make use of the modulo operator, which allows us to rewrite the above as following:

$$f(x,y) = f([x]_\Omega, [y]_\Omega)$$

Such that, $[x]_\Omega = x \bmod \Omega$.

Now, we define the discrete Radon transform as follows for this given setup (Where, t is defined on the discrete domain of 0 to $\Omega - 1$.):

$$R_D f(P,t) = \sum_{\substack{y=0 \\ y=tx+}}^{\Omega-1} \sum_{x=0}^{\Omega-1} f(x,y) = \sum_{x=0}^{\Omega-1} f(x,[tx+P]_\Omega)$$

Similarly, we can define a horizontal and a vertical discrete Radon transform as well. Let us denote $R_D f(P,t) = \tilde{f}(P,t)$ and let $\tilde{f}_V(P)$ and $\tilde{f}_H(P)$ be vertical and horizontal discrete Radon transforms respectively.

$$\tilde{f}_V(P,t) = \sum_{y=0}^{\Omega-1} f(t,y) \; \& \; \tilde{f}_H(P,t) = \sum_{x=0}^{\Omega-1} f(x,t)$$

Reverting back to our primary formulation, we can discretize this further by assuming (x,y) are grid points on a rectangular system. Where, x lies on the discrete domain of 0 to $n-1$ and y lies on the discrete domain of 0 to $m-1$. We will be taking the discrete Radon transform of the following modified function assisted by the product of unit functions:

$$R_D\{u(n-1-x)u(x)u(m-1-y)u(y)f(x,y)\}(P,t) = \sum_{\substack{x=0 \\ U^T X=P}}^{n-1} \sum_{y=0}^{m-1} f(x,y)$$

Where, the assumed definition of the unit function is:

$$u(x) = \begin{cases} 1; x \geq 0 \\ 0; x < 0 \end{cases}$$

Thus, a new definition for discrete Radon transforms is written as:

$$R_D^{(n,m)} f(P,t) = \sum_{x=0}^{n-1} \sum_{y=0}^{m-1} f(x,y)\delta(P - x\cos t - y\sin t)$$

Example 9.1: Assume that $f(x,y) > 0$ and $\varepsilon > 0$. Prove or disprove that $R_D^\varepsilon f(P,t) = \sum_{x,y\in\mathbb{Z}^2} f(x,y)M_\varepsilon(x,y)$ (as defined earlier on) is greater than $R_D f(P,t) = \sum_{x,y\in\mathbb{Z}^2} f(x,y)M(x,y)$.

As we had defined previously, $R_D^{\varepsilon} f(P,t)$ can be rewritten using $M(x,y)$.

$$R_D^{\varepsilon} f(P,t) = \sum_{x,y \in \mathbb{Z}^2} f(x,y) M_{\varepsilon}(x,y)$$

Let, $M_{\varepsilon}(x,y) = M(x,y) + M^*(x,y)$. Such that, M^* is defined as follows:

$$M^*(x,y) = \begin{cases} 1\,;\, |x\cos t + y\sin t - P| \leq \varepsilon \,/\, \{0\} \\ 0\,;\, |x\cos t + y\sin t - P| > \varepsilon \end{cases}$$

$\varepsilon \,/\, \{0\}$ simply implies considering all the cases except the case when the function's evaluation results in $\mathbf{0}$.

$$R_D^{\varepsilon} f(P,t) = \sum_{x,y \in \mathbb{Z}^2} f(x,y) M(x,y) + \sum_{x,y \in \mathbb{Z}^2} f(x,y) M^*(x,y)$$

But as we already know that $f(x,y) > \mathbf{0}$ and both $M^*(x,y), M(x,y) \geq \mathbf{0}$ for any given $x, y \in \mathbb{Z}^2$, we can confidently assert the following:

$$R_D^{\varepsilon} f(P,t) > R_D f(P,t)$$

Moving ahead, as the discrete Radon transform is naturally connected to the standard Radon transform, we can state the Fourier slice theorem in this context too. Recall the following identity:

$$R\left\{\sum_{m,n \in \mathbb{Z}^2} \delta(x-n, y-m) f(x,y)\right\}(P,U) = R_D f(P,U)$$

We simply replace the above identity in the standard Fourier slice theorem identity to achieve a discrete-centric identity:

$$F_y\{R_D f(y,U)\}(s) = F\left\{\sum_{m,n \in \mathbb{Z}^2} \delta(x-n, y-m) f(x,y)\right\}(sU)$$

Here, the RHS of the expression readily evaluates to the following:

$$F_y\{R_D f(y,u,v)\}(s) = \sum_{m,n \in \mathbb{Z}^2} e^{-isum - isv}\; f(m,n)$$

By substituting $(u,v) = (\cos t\,, \sin t)$, we obtain the following result:

$$F_y\{R_D f(y,t)\}(s) = \sum_{m,n \in \mathbb{Z}^2} e^{-ismcost\, - isnsint} f(m,n)$$

The above is called as the discretized Fourier-slice theorem. So, to simply state, let us assume $R_D f(P, U) = Rh(P, u)$ (Where, h is some appropriate function to satisfy the criterion), then the discretized Fourier-slice theorem can be constructed as follows. Firstly, stating the standard Fourier-slice identity:

$$F_y\{Rh(y, U)\}(s) = F\{h\}(sU)$$

Thus, the discretized Fourier-slice identity will be:

$$F_y\{R_D f(y, U)\}(s) = F\{h\}(sU)$$

Moving forward, we discuss the discretized version of the adjoint Radon transform. To construct that, we need to firstly restate its formal definition and appropriate Riemann sum approximation. Assume, T to be some natural number, (x_m, y_n) to be a point on the discretized rectangular grid and $Kf(x, y) = f^*(x, y)$. Firstly, stating the integral-based definition of adjoint Radon transforms using a filter operator K in two-dimensions

$$R^*Kf(x, y) = R^*f^*(x, y) = \int_0^\pi f^*(x\,cos\,t + y\,sin\,t\,, t)\,dt = f(x, y)$$

Thus, $f(x, y)$ can be written as such using the adjoint Radon transform definition:

$$f(x, y) = \int_0^\pi f^*(x\,cos\,t + y\,sin\,t\,, t)\,dt$$

By using the Riemann sum formula, we can approximate $f(x_n, y_m)$ with relative ease now:

$$f(x_n, y_m) \approx \sum_{\tau=0}^{T-1} f^*(x_n\,cos\,t_\tau + y_m\,sin\,t_\tau\,, t_\tau)\Delta\tau$$

Here, t_τ is generalized arbitrary method of measurement, depends entirely on the given context and the mode of preference of the application. Therefore, one possible extrapolation of the discretized adjoint Radon transform can be stated as follows:

$$R_D^* f(x, y) = \sum_{\tau=0}^{T-1} f^*(x\,cos\,\tau + y\,sin\,\tau\,, \tau)$$

Coming back to discrete Radon transforms, let us discuss one last important example here.

Example 9.2: Assume that the line constraint was given to be $x = sy + P$, develop a valid discretization scheme and an apt mathematical formulation for discrete Radon transform of $f(x, y)$.

From what we already know, we define integer parameters λ, μ and Ω be some prime as x, y are defined on the discrete domain of 0 to $\Omega - 1$.

$$f(x, y) = f(\lambda x + \Omega, \mu y + \Omega)$$

This enables us to imply the following result:

$$f(x, y) = f([x]_\Omega, [y]_\Omega)$$

Ultimately, $R_D f(P, s)$ is expressed using the following summation notation:

$$R_D f(P, s) = \sum_{y=0}^{\Omega-1} \sum_{\substack{x=0 \\ x=sy+P}}^{\Omega-1} f(x, y)$$

As we are given that $x = sy + P$, we can use the modulo operator to the state the final result:

$$R_D f(P, s) = \sum_{y=0}^{\Omega-1} f([sy + P]_\Omega, y)$$

This concludes our discussion of discrete Radon transforms and our last topic of discussion will be finite Radon transforms in which we will be exploring the algebraic machinery that goes into work.

Finite Radon transforms

We will start by defining all the required preliminaries. Let us define X to be the input set with n elements ($|X| = n$), Y to be the subset of all possible subsets of X ($Y \subseteq P(X)$) and it has a cardinality of m elements such that $m \geq n$, x to be some arbitrary element of given sets X & y to be a valid subset of Y, a function $f : X \to \mathbb{C}$ and ultimately $Rf : C(X) \to C(Y)$ such that C denotes complex valued functions. Using all this, we can define a finite Radon transform as follows:

$$Rf(y) = \sum_{x \in y} f(x)$$

We will go through a lot of examples in this section to clearly understand the underlying algebra. Let us work out the first example of this section.

<u>Example 10.1:</u> For given $X = \{a, b, c, d\}$ (Such that $a, b, c, d \in \mathbb{R}$), $Y = \{(a, b), (b, d), (a, d)\}$ and $f(x) = x$ calculate (i) $Rf(a, b)$ (ii) $Rf(b, d)$ (iii) $Rf(a, d)$ (iv) $\sum_{y \in Y} Rf(y)$

Computing the value of $Rf(a, b)$ by using the definition itself.

$$Rf(a, b) = \sum_{x \in (a,b)} f(x) = f(a) + f(b) = a + b$$

Similarly, calculating the value of $Rf(b, d)$.

$$Rf(b, d) = \sum_{x \in (b,d)} f(x) = f(b) + f(d) = b + d$$

The result obtained after the calculations of $Rf(a, d)$ is not quite different either.

$$Rf(a, d) = \sum_{x \in (a,d)} f(x) = f(a) + f(d) = a + d$$

To obtain the solution of (iv), we simply add up all our previous results.

$$\sum_{y \in Y} Rf(y) = Rf(a, b) + Rf(b, d) + Rf(a, d) = 2a + 2b + 2d$$

That gives us a basic gist of finite Radon transforms. To proceed further, let us understand few important structures of X, Y.

Let us assume, $X = \{u, v, w\}$ and $Y = \{(u, v), (u, w), (v, w)\}$. Notice that if we take all elements in X as vertices of the graph then Y will represent all the connecting edges present in the graph. In this case, the resulting graph obtained is a triangle and this graph is often called as K_3 in graph theory.

Similarly, for $X = \{u, v, w, p\}$ and $Y = \{(u, v), (u, w), (u, p), (v, w), (v, p), (w, p)\}$, we obtain a graph of a quadrilateral with its diagonals connected, which is represented by K_4 in graph theory.

In general, for $|X| = n$ (such that $n \geq 2$) we have $|Y| = \binom{n}{2}$ to encompass a K_n graph.

On the contrary, let us assume $X = \{u, v, w, p\}$ and $Y = \{(u, v), (v, w), (w, p), (p, u)\}$. From this we obtain only the perimeter of the quadrilateral and thus is represented as a cyclic group of order 4 denoted mathematically as C_4. The same can also be represented using dihedral group of order 2 given as $D_2 = \{i, \mu, \rho, \mu\rho\}$.

In general, for $|X| = n$ (such that $n \geq 1$) we have $|Y| = n$ to encompass a C_n graph.

The direct correlation between finite Radon transforms and graph theory, suggests that we can represent the transform itself as a point-incidence matrix based on X, Y. Let us define the point-incidence matrix as follows:

$$A(i, j) = \begin{cases} 1 \,; \; x_j \in y_i \\ 0 \,; \; x_j \notin y_i \end{cases}$$

Such that, $\forall x_j \in X$ and $\forall y_i \in Y$ as $1 \leq j \leq n$ and $1 \leq i \leq m$. So basically, rows are represented by subsets of Y, columns are represented by the elements of X and $dim(A) = m \times n$.

Now, we assume $f(x) = \left[f\left(x_j\right)\right]$ and it becomes obvious that $dim\left(f(x)\right) = n \times 1$. Because of which we have:

$$Rf(y) = A_{(m,n)} f_{(n,1)} = (Af)_{(m,1)}$$

Which ultimately tells us that, finite Radon transform is simply a matrix multiplication in disguise.

Let us solve an example to further understand this concept.

<u>Example 10.2:</u> It is given that $X = \{u, v, w, p\}$ (Such that $u, v, w, p \in \mathbb{R}$),$Y = \{y_1, y_2, y_3\}$ as $y_1 = (u, v, w)$, $y_2 = (u, v, p)$, $y_3 = (w, p)$ and $f(x) = e^x$. Compute (i) the point-incidence matrix A and (ii) the finite Radon transform $Rf(y)$ in matrix form such that $\forall y \in Y$.

Firstly, calculating the point-incidence matrix A by observing the present elements in y_1, y_2, y_3 and marking their presence by either 0 or 1.

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Now, f is stated as follows:

$$f = \begin{bmatrix} f(u) \\ f(v) \\ f(w) \\ f(p) \end{bmatrix} = \begin{bmatrix} e^u \\ e^v \\ e^w \\ e^p \end{bmatrix}$$

Thus, the finite Radon transform $Rf(y)$ is computed as such:

$$Rf(y) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} e^u \\ e^v \\ e^w \\ e^p \end{bmatrix} = \begin{bmatrix} e^{u+v+w} \\ e^{u+v+p} \\ e^{w+p} \end{bmatrix}$$

That is our final answer. Notice that all individual results of $Rf(y_1), Rf(y_2), Rf(y_3)$ are stacked in a matrix form here.

A brief introduction to cyclic integer groups is crucial as it allows us to deal with more applications. A cyclic integer group of order n (where n is some positive integer greater than 1), defined as $\mathbb{Z}_n = \{i | i \in \mathbb{N} \cup \{0\}, i \leq n-1\}$ has the following properties:

$$a +_n b = (a+b) \, mod \, n$$

Such that, $a, b \in \mathbb{Z}_n$. To understand this definition more intuitively, take a quick example with $a = n-2$ and $b = 3$, Here $a +_n b = (n+1) \, mod \, n = 1$. Now, note that $+_n$ sign is dropped when working in a given group/ring $\mathbb{Z}_n$ as it is automatically understood.

To be fair, $\mathbb{Z}_n$ is used in all of the computations as a ring as it inhibits both addition and multiplication, denoted as $\langle \mathbb{Z}_n, +, \times \rangle$. The multiplication operation in $\mathbb{Z}_n$ works as such assuming as per usual $a, b \in \mathbb{Z}_n$.

$$a *_n b = (ab) \, mod \, n$$

Now let us solve a finite Radon transform problem using our newly acquired knowledge.

Example 10.3: It is given that $n \geq 10$, $X = \mathbb{Z}_n$, $Y = \mathbb{Z}_n^2$ and $f(x) = x + 1$. Calculate the finite Radon transform of $Rf(n-5,4)$ and $Rf(n-3,4)$.

Simply using the definition to compute the required transforms.

$$Rf(n-5,4) = \sum_{x \in (n-5,4)} f(x) = f(n-5) + f(4) = n - 4 + 5 = n + 1$$

But in $\mathbb{Z}_n$, we already know that $n + 1 = (n+1)\, mod\, n = 1$. Thus, we have:

$$Rf(n-5,4) = 1$$

Similarly, following the same process, for $Rf(n-3,4)$ we have:

$$Rf(n-3,4) = \sum_{x \in (n-3,4)} f(x) = f(n-3) + f(4) = (n+3)\, mod\, n = 3$$

If the reader is curious, they should rework the same given example but with $f(x) = x^2 + x + 1$, $X = \mathbb{Z}_5$ and $Y = \mathbb{Z}_5^2$ instead.

Alright, with that done, we can now explore the injectivity observed in Radon transforms. Before diving into the formalities, let us explore injectivity conditions of finite Radon transforms.

Primarily, we define $G_x = \{y \in Y | x \in y\}$, which implies G is the set of all the subsets of Y such that they all definitively contain the element x given $\forall x \in X$.

To qualitatively understand this, let us work through an example.

Example 10.4: It is given that $X = \{u, v, w, p\}$ (Such that $u, v, w, p \in \mathbb{R}$) and $Y = \{U, V, W, P\}$ such that $U = (u, v)$, $V = (v, w)$, $W = (w, p)$ and $P = (p, u)$. Calculate G_u, G_v, G_w, G_p.

On recalling the original definition, we have:

$G_u = \{y \in Y | u \in y\} = \{(u, v), (p, u)\} = \{U, P\}$

$G_v = \{y \in Y | v \in y\} = \{(u, v), (v, w)\} = \{U, V\}$

$G_w = \{y \in Y | w \in y\} = \{(v, w), (w, p)\} = \{V, W\}$

$G_p = \{y \in Y | p \in y\} = \{(w, p), (p, u)\} = \{W, P\}$

Notice that, as $Y \cong C_4$ the resulting cardinalities are $|G_u| = |G_v| = |G_w| = |G_p| = 2$ and another set of nice conclusions being, $|G_u \cap G_v| = |G_u \cap G_p| =$

$|G_v \cap G_w| = 1$ but $|G_u \cap G_w| = |G_v \cap G_p| = 0$. We will learn later how this is problematic and can lead to injectivity-based issues while calculating finite Radon transforms.

Now that we firmly understand the role of the injectivity set G, we can formally define the conditions for the existence of injectivity on Radon transforms. Let, a, b be some positive integers such that $a \neq b$. We also assume, $\forall \mu, \nu \in X$ and the injectivity set G is defined for a said μ as, $G_\mu = \{y \in Y | \mu \in y\}$. Based on these assumptions, the following equations must hold true for R to be injective.

$$|G_\mu| = a$$

$$|G_\mu \cap G_\nu| = b$$

Now going back to our previous example, we specifically found out that $|G_u \cap G_w| = |G_v \cap G_p| = 0$, which directly violates the above set constraints due to which for some given viable $f(x)$ in this case, Rf is not injective by nature. This directly affects the process of reconstructing f from Rf due to which this finite inverse Radon transform will not exist in many cases unless changes are brought to both X, Y.

Anyway, let us now look at a few examples in calculating finite inverse Radon transform.

<u>Example 10.4:</u> Assume that $X = \{a, b, u, v\}$ (Such that $a, b, u, v \in \mathbb{R}$) and $Y = \{y_i | 1 \leq i \leq 6\}$ such that $y_1 = \{a, b\}$, $y_2 = \{a, u\}$, $y_3 = \{a, v\}$, $y_4 = \{b, u\}$, $y_5 = \{b, v\}$ and $y_6 = \{u, v\}$. Suppose, it is known to us that $Rf(y) = \begin{bmatrix} a+b+\beta \\ a+u+\beta \\ a+v+\beta \\ b+u+\beta \\ b+v+\beta \\ u+v+\beta \end{bmatrix}$ such that $\beta \in \mathbb{R}$, find out the value of $f(x)$. (As per usual assuming $x \in X, y \in Y$)

Firstly, notice that earlier mentioned injectivity constraints are not violated. Following the defined injectivity set G, let us observe the resulting cardinalities.

$G_a = \{y_1, y_2, y_3\}$ ➔ $|G_a| = 3$, $G_b = \{y_1, y_4, y_5\}$ ➔ $|G_b| = 3$,

$G_u = \{y_2, y_4, y_6\} \Rightarrow |G_u| = 3$, $G_v = \{y_3, y_5, y_6\} \Rightarrow |G_v| = 3$

The first condition has been satisfied, now let us check the second condition

$|G_a \cap G_b| = 1$, $|G_a \cap G_u| = 1$, $|G_a \cap G_v| = 1$, $|G_b \cap G_u| = 1$, $|G_b \cap G_v| = 1$, $|G_u \cap G_v| = 1$

As one can see, both the conditions are well satisfied due to which there must be no issues in calculating the inverse Radon transform here. We simply use the Radon transform definition and compare the given values.

$$Rf(y) = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} f(a) \\ f(b) \\ f(u) \\ f(v) \end{bmatrix} = \begin{bmatrix} f(a) + f(b) \\ f(a) + f(u) \\ f(a) + f(v) \\ f(b) + f(u) \\ f(b) + f(v) \\ f(u) + f(v) \end{bmatrix}$$

On comparing the two, it becomes quite easy to recognize that all these are symmetric functional equations. Let us consider one of these, and obtain a valid value of $f(x)$.

$$f(a) + f(b) = a + b + \beta$$

Many methods exist to solve the above given functional equation but we will be using the simplest one in this case, assuming the Taylor series polynomial ansatz of $f(x) = Ax + B$ such that $A, B \in \mathbb{R}$ and then substituting them back into the original functional equation to find the necessary coefficients.

$$Aa + B + Ab + B = a + b + \beta$$

$$A(a + b) + 2B = a + b + \beta$$

By comparing, we have $A = 1, B = \frac{\beta}{2}$ because of which $f(x)$ can be given as:

$$f(x) = x + \frac{\beta}{2}$$

We have covered almost everything related to Radon transforms in general now but before we end the segment, let us solve one more inverse of finite Radon transform problem but with Cauchy's functional equation twist.

Example 10.5: It is given that $X = \{u, v, w\}$ and $Y = \{y_1, y_2, y_3\}$ such that $y_1 = \{u, w\}$, $y_2 = \{v, w\}$ and $y_3 = \{u, v\}$. Suppose, we know that $Rf(y) = \begin{bmatrix} f(uw) \\ f(vw) \\ f(uv) \end{bmatrix}$, find an appropriate value of $f(x)$. (As per usual assuming $x \in X, y \in Y$)

It is not a necessity to check whether the injectivity constraints are satisfied but as Y is a cyclic group yet again, the result is quite straight-forward.

$$|G_u| = |G_v| = |G_w| = 2$$

$$|G_u \cap G_v| = |G_u \cap G_w| = |G_v \cap G_w| = 1$$

As always, representing the Radon transform using point-incidence matrix.

$$Rf(y) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} f(u) \\ f(v) \\ f(w) \end{bmatrix} = \begin{bmatrix} f(u) + f(w) \\ f(v) + f(w) \\ f(u) + f(v) \end{bmatrix}$$

Notice that we obtain a system of symmetric functional equations yet again and by comparison, we obtain the following sample functional equation.

$$f(u) + f(w) = f(uw)$$

Yet again, many ways exist to solve the given Cauchy's functional equation but as our book honours integral transforms, we will be using Mellin transform to solve this functional equation and assume implicitly that $f(x)$ has a valid Mellin transform.

Let, $e^{f(x)} = F(x)$ because of which we can rewrite the original functional equation as follows:

$$e^{f(u)+f(w)} = e^{f(uw)} \Rightarrow F(u)F(w) = F(uw)$$

Now, Mellin transform w.r.t u is taken on both sides which leads to the following (If the reader is unfamiliar with Mellin transforms and its properties, refer to the Appendix where all the supplementary topics are discussed):

$$M\{F(u)F(w)\}(s) = M\{F(uw)\}(s)$$

$$F(w)M\{F(u)\}(s) = w^{-s}M\{F(u)\}(s)$$

Cancelling $M\{F(u)\}(s)$ on both sides and substituting some constant $-c$ in place of transform parameter s.

$$F(w) = w^c \Rightarrow f(w) = c\,ln(w)$$

Thus, the final value of $f(x)$ in this given problem is stated as:

$$f(x) = c\,ln(x)$$

This concludes the discussion of the theory of Radon transforms, if the reader is further interested in learning more about Radon transforms, I'd highly recommend reading research articles on the technology used in tomography and CT scans and other than that, algorithmic approaches to discrete Radon transforms can also help the curious reader a lot.

Appendix

Starting with basic preliminaries, the integral definition of gamma function is given as such:

$$\Gamma(p) = \int_0^\infty e^{-x} x^{p-1}\,dx$$

A very Famous known value of the gamma function is:

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Because of which, the Gaussian integral can be reinterpreted and written as follows:

$$\int_0^\infty e^{-x^2}\,dx = \int_0^\infty e^{-x} x^{-\frac{1}{2}}\,dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Using the same logic, we can extend the same result to n-dimensions:

$$\int_0^\infty \int_0^\infty \cdots \int_0^\infty e^{-x_1^2-x_2^2-\cdots-x_n^2}\,dx_1\,dx_2 \ldots dx_n = \Gamma^n\left(\frac{1}{2}\right) = \pi^{\frac{n}{2}}$$

Next up, we discuss about the δ function. It is defined as follows:

$$\delta(x) = \begin{cases} \infty\,;\, x = 0 \\ 0\,;\, x \in \mathbb{R} \,/\, \{0\} \end{cases}$$

The following are two very important identities of the δ function.

(a) $\int_{x\in\mathbb{R}} \delta(x)\, dx = 1$
(b) $\int_{x\in\mathbb{R}} f(x)\delta(x-a)\, dx = f(a)$ (Assuming, a is some real constant and f is a well-defined function)

Moving forward, we discuss about the Laplace transforms. A Laplace transform is defined as follows for a suitable f.

$$L\{f(x)\}(s) = \int_0^\infty e^{-sx} f(x)\, dx$$

Key identities of Laplace transforms are:

(1) $L\{a\}(s) = a$ (Where, a is some real constant)
(2) $L\{\delta(t-a)\}(s) = e^{-sa}$
(3) $L\{x^n\}(s) = \frac{\Gamma(n+1)}{s^{n+1}}$
(4) $L\{e^{-ax}\}(s) = \frac{1}{s+a}$
(5) $L\{sin(at)\}(s) = \frac{a}{s^2+a^2}$
(6) $L\{cos(at)\}(s) = \frac{s}{s^2+a^2}$

Key properties of Laplace transforms are (Assuming, $L\{f(x)\}(s) = F(s)$)

(1) $L\{f(ax)\}(s) = \frac{1}{a}F\left(\frac{s}{a}\right)$
(2) $L\{e^{ax}f(x)\}(s) = F(s-a)$
(3) $L\left\{\frac{f(x)}{x}\right\}(s) = \int_s^\infty F(t)\, dt$
(4) $L\{f'(x)\}(s) = sF(s) - f(0)$
(5) $L\{f''(x)\}(s) = s^2F(s) - sf(0) - f'(0)$
(6) $L\{f^{(n)}(x)\}(s) = s^nF(s) - \sum_{i=1}^{n} s^{n-i} f^{(i-1)}(0)$

Also, the convolution in Laplace transforms is given as (As per usual, assuming f, g to be two suitable functions):

$$f * g = \int_0^x f(t)g(x-t)\, dt$$

$$L\{f(x) * g(x)\}(s) = L\{f(x)\}(s)L\{g(x)\}(s)$$

Whereas, the inverse Laplace transform integral formula is given as follows (Assuming, $L\{f(x)\}(s) = F(s)$):

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{sx}\, ds$$

Now, moving on to discuss the Fourier transforms. Firstly, Fourier series for a suitable function $f(x)$ is given as follows:

$$f(x) = \frac{a_0}{2} + \sum_{n\in\mathbb{N}} a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)$$

Such that, $a_n = \frac{2}{L}\int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$ and $b_n = \frac{2}{L}\int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx$. Based on this, Fourier sine and cosine transforms are stated as follows:

$$F_s\{f(x)\}(n) = \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \tilde{f}_s(n)$$

$$F_c\{f(x)\}(n) = \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \tilde{f}_c(n)$$

Key properties include:

(1) $F_s\{f'(x)\} = -\left(\frac{n\pi}{a}\right)\tilde{f}_c(n)$

(2) $F_s\{f''(x)\}(n) = -\left(\frac{n\pi}{a}\right)^2 \tilde{f}_s(n)$ (Assuming, $f(0) = f(L) = 0$)

(3) $F_c\{f'(x)\}(n) = \left(\frac{n\pi}{a}\right)\tilde{f}_s(n)$ (Assuming, $f(0) = f(L) = 0$)

(4) $F_c\{f''(x)\}(n) = -\left(\frac{n\pi}{a}\right)^2 \tilde{f}_c(n)$ (Assuming, $f(0) = f(L) = 0$)

Inverses of the Fourier sine and cosine transforms are nothing but Fourier sine and cosine series. Consequently, Fourier transform as a whole can be defined as follows:

$$F\{f(x)\}(k) = \int_{x\in\mathbb{R}} e^{-ikx} f(x)\, dx = \tilde{f}(k)$$

Fourier inverse of which is given as:

$$f(x) = \int_{k\in\mathbb{R}} e^{ikx}\tilde{f}(k)\,dk$$

Properties of Fourier transforms:

(1) $F\{f(x+a)\}(k) = e^{ika}\tilde{f}(k)$
(2) $F\{e^{iax}f(x)\}(k) = \tilde{f}(k-a)$
(3) $F\{f(ax)\}(k) = \frac{1}{a}\tilde{f}\left(\frac{k}{a}\right)$
(4) $F\{f^{(n)}(x)\}(k) = (ik)^n\tilde{f}(k)$ (Assuming, $f^{(i)}(0) = 0\ \forall 0 \leq i \leq n$)

Meanwhile, convolution in Fourier transforms is given as follows (As per usual, assuming f, g are suitable functions):

$$f(x) * g(x) = \int_{t\in\mathbb{R}} f(t)g(x-t)\,dt$$

$$F\{f(x) * g(x)\}(k) = F\{f(x)\}(k)F\{g(x)\}(k)$$

Moving ahead, the definition of Mellin transforms is given by the following integral definition.

$$M\{f(x)\}(p) = \int_0^\infty x^{p-1}f(x)\,dx = \psi(p)$$

Key properties include (Assuming, $M\{f(x)\}(P) = \psi(p)$):

(1) $M\{f(ax)\}(p) = a^{-p}\psi(P)$
(2) $M\{x^a f(x)\}(p) = \psi(p+a)$
(3) $M\{f(x^\rho)\}(p) = \frac{1}{\rho}\psi\left(\frac{p}{\rho}\right)$ (Assume, $\rho \in \mathbb{N}$)
(4) $M\{f'(x)\}(p) = -(s-1)\psi(s-1)$

Convolution in Mellin transforms is given as:

$$f(x) *_M g(x) = \int_0^\infty f(t)g\left(\frac{x}{t}\right)\frac{dt}{t}$$

$$M\{f(x) * g(x)\}(p) = M\{f(x)\}(p)M\{g(x)\}(p)$$

Whereas, the inverse Mellin transform is given by:

$$f(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} x^{-p} \psi(p) \, dp$$

The next transform we explore is the Hilbert transform, defined by the following integral formulation:

$$H\{f(x)\}(t) = \frac{1}{\pi} \int_{x \in \mathbb{R}} \frac{f(x)}{x-t} dx = f_H(t)$$

Key properties of Hilbert transform include:

(1) $H\{f(x+a)\}(t) = f_H(t+a)$
(2) $H\{f(ax)\}(t) = sgn(a) f_H(at)$
(3) $H\{f(ax)\}(t) = f_H(at)$
(4) $H\{f'(x)\}(t) = \frac{d}{dt} f_H(t)$

Whereas, the Inverse Hilbert transform is given as:

$$f(x) = \frac{1}{\pi} \int_{t \in \mathbb{R}} \frac{f_H(t)}{t-x} dt$$

With that, we have discussed all the important transforms and properties, let us now discuss the heat equation PDE. The heat equation in one-dimension is given as:

$$u_t = c u_{xx}$$

Such that, $c > 0$. Similarly, for two-dimensions, we have:

$$u_t = c u_{xx} + c u_{yy}$$

Ultimately, for n-dimensions we assume $X = [x_i] \; \forall i \leq n$, we can say:

$$u_t = c \sum_{i=1}^{n} u_{x_i x_i} = c\Delta^2 u = c\nabla u$$

General solution for heat equation in n-dimensions is given as follows:

$$u(x,t) = \left(\frac{1}{(4\pi ct)^{n/2}} e^{-\frac{\|X\|^2}{4ct}} \right) * u(x,0) = \int_{T \in \mathbb{R}} G(T-X, t) u(T, 0) \, dT$$

Such that the heat kernel $G(x,t) = \frac{1}{(4\pi ct)^{n/2}} e^{-\frac{\|X\|^2}{4ct}}$.

Similarly, if we assume only the one-dimensional case with the given boundary conditions and initial conditions: $u(0,t) = u(a,t) = 0$ & $u(x,0) = f(x)$, we obtain the following result:

$$u_t = cu_{xx}$$

Applying Fourier sine transform on both sides of the equation (Assuming, $F_s\{u(xt)\}(n) = \tilde{v}(n,t)$)

$$F_s\{u_t(x,t)\}(n) = cF_s\{u_{xx}(x,t)\}(n)$$

$$\tilde{v}_t(n,t) = -c\left(\frac{n\pi}{a}\right)^2 \tilde{v}(n,t)$$

Solving the above ODE by introducing $e^{l(t)}$ on both sides of the expression:

$$e^{l(t)}\tilde{v}_t(n,t) - c\left(\frac{n\pi}{a}\right)^2 e^{l(t)}\tilde{v}(n,t) = 0$$

Setting $l(t) = -c\left(\frac{n\pi}{a}\right)^2 t$, to reduce the preceding equation as follows:

$$\frac{d}{dt}\left(e^{-c\left(\frac{n\pi}{a}\right)^2 t}\tilde{v}(n,t)\right) = 0 \rightarrow \tilde{v}(n,t) = q(n)e^{-c\left(\frac{n\pi}{a}\right)^2 t}$$

But, $\tilde{v}(n,0) = q(n)$ and $F_s\{u(x,0)\}(n) = F_s\{f(x)\}(n)$ which basically implies $\tilde{v}(n,0) = \tilde{f}_s(n)$. Thus, $q(n) = \tilde{f}_s(n)$:

$$\tilde{v}(n,t) = \tilde{f}_s(n)e^{-c\left(\frac{n\pi}{a}\right)^2 t}$$

Taking inverse on both sides results in the following Fourier sine transform:

$$u(x,t) = \frac{2}{a}\sum_{n\in\mathbb{N}} \tilde{f}_s(n)e^{-c\left(\frac{n\pi}{a}\right)^2 t} sin\left(\frac{n\pi x}{a}\right)$$

This is the solution to 1D heat equation under Dirichlet conditions. The same trick of Fourier sine transform can be used to solve many such linear PDEs under Dirichlet or Neumann conditions. Let us now, assume the initial and boundary conditions to be: $u(x,0) = f(x)$, $u_t(x,0) = g(x)$ & $u(0,t) = u(a,t) = 0$ to solve the following PDE:

$$u_{tt} = \beta u_{xx}$$

Such that, $\beta > 0$. Firstly, applying Fourier sine transform on both sides and also, assume that $F_s\{u(x,t)\}(n) = \tilde{v}(n,t)$:

$$\tilde{v}_{tt}(n,t) + \beta\left(\frac{n\pi}{a}\right)^2 \tilde{v}(n,t) = 0$$

The solution to the preceding ODE can be stated as follows:

$$\tilde{v}(n,t) = A_n \sin\left(\frac{n\pi\sqrt{\beta}}{a}t\right) + B_n \cos\left(\frac{n\pi\sqrt{\beta}}{a}t\right)$$

Now by using the initial value conditions, we can assume that $\tilde{v}(n,0) = F_s\{u(x,0)\}(n) = F_s\{f(x)\}(n) = \tilde{f}_s(n)$ and $\tilde{v}_t(n,0) = F_s\{u_t(x,0)\}(n) = F_s\{g(x)\}(n) = \tilde{g}_s(n)$. Thus A_n, B_n can be determined as follows:

$$\tilde{v}(n,t) = A_n \sin\left(\frac{n\pi\sqrt{\beta}}{a}t\right) + B_n \cos\left(\frac{n\pi\sqrt{\beta}}{a}t\right)$$

$$\tilde{v}_t(n,t) = A_n\left(\frac{n\pi\sqrt{\beta}}{a}\right)\cos\left(\frac{n\pi\sqrt{\beta}}{a}t\right) - B_n\left(\frac{n\pi\sqrt{\beta}}{a}\right)\sin\left(\frac{n\pi\sqrt{\beta}}{a}t\right)$$

Substituting $t = 0$ in the above equations gives us the following:

$$A_n = \tilde{f}_s(n) \;\&\; B_n = \frac{a\tilde{g}_s(n)}{n\pi\sqrt{\beta}}$$

Taking the inverse transform on $\tilde{v}(n,t)$, gives us the final solution:

$$u(x,t) = \frac{2}{a}\sum_{n\in\mathbb{N}}\left(\tilde{f}_s(n)\cos\left(\frac{n\pi\sqrt{\beta}t}{a}\right) + \frac{a\tilde{g}_s(n)}{n\pi\sqrt{\beta}}\sin\left(\frac{n\pi\sqrt{\beta}t}{a}\right)\right)$$

The same wave equation can be solved generally as follows:

$$u_{tt} - \beta u_{xx} = 0$$

$$(D_t^2 - \beta D_x^2)u(x,t) = 0 \rightarrow (D_t - \sqrt{\beta}D_x)(D_t + \sqrt{\beta}D_x)u(x,t) = 0$$

By factoring, we obtain two transport PDEs and the sum of these two solutions will be the general solution. Let us solve both of these systematically:

$$u_t - \sqrt{\beta}u_x = 0$$

Let, $E = x - \sqrt{\beta}t$ and $T = x + \sqrt{\beta}t$ changing the coordinate system to (E, T) instead. Thus, derivative computations will be:

$$u_t = u_E\left(\frac{dE}{dt}\right) + u_T\left(\frac{dT}{dt}\right) = -\sqrt{\beta}u_E + \sqrt{\beta}u_T$$

$$\sqrt{\beta}u_x = \sqrt{\beta}u_E\left(\frac{dE}{dx}\right) + \sqrt{\beta}u_T\left(\frac{dT}{dx}\right) = \sqrt{\beta}u_E + \sqrt{\beta}u_T$$

Therefore, thanks to this diagonalization, our original PDE reduces to the following (Let, P be some well-defined analytic function):

$$u_T = 0 \Rightarrow u(x,t) = P\left(x + \sqrt{\beta}t\right)$$

Analogously, we solve the second transport type PDE with the same coordinate changes and the equation ultimately reduces to (Let, Q be some well-defined analytic function):

$$u_E = 0 \Rightarrow u(x,t) = Q\left(x - \sqrt{\beta}t\right)$$

Thus, the general solution can be stated as follows:

$$u(x,t) = P\left(x + \sqrt{\beta}t\right) + Q\left(x - \sqrt{\beta}t\right)$$

Similarly, for initial conditions $u(x,0) = f(x)$ and $u_t(x,0) = g(x)$, the above solution transforms into the following:

$$u(x,t) = \frac{1}{2}\left[f\left(x + \sqrt{\beta}t\right) + f\left(x - \sqrt{\beta}t\right)\right] + \frac{1}{2\sqrt{\beta}}\int_{x-\sqrt{\beta}t}^{x+\sqrt{\beta}t} g(y)\,dy$$

Let us also learn solving functional equations using integral transforms we have learnt so far. Assume h to be some well-behaved function under Mellin transforms, we are tasked to solve the following functional equation:

$$h(xy) = h(x)h(y)$$

Assume, $M\{h(x)\}(p) = H(p)$. By making use of this assumption, we take Mellin transform on both sides:

$$M_x\{h(xy)\}(p) = h(y)M\{h(x)\}(p)$$

$$y^{-p}H(p) = h(y)H(p)$$

Cancelling $H(p)$ on both sides and assuming the transform parameter p to be replaced with some real constant $-c$, gives us the following:

$$h(y) = y^c$$

Next up, using the same function h, we will be attempting to solve the following functional equation:

$$h(x) + h(y) = h(x + y)$$

Let $e^{h(x)} = f(x)$ and by using this we exponentially raise the original functional equation: ➔

$$e^{h(x)}e^{h(y)} = e^{h(x+y)} \text{➔} f(x)f(y) = f(x + y)$$

Assuming, $M\{f(x)\}(p) = F(p)$ and taking Mellin transform on both sides:

$$f(y)M\{f(x)\}(p) = M_x\{f(x + y)\}(p)$$

$$f(y)F(P) = y^{-p}F(p)$$

Cancelling $F(p)$ on both sides and replacing the transform parameter p with some real constant $-c$ and then ultimately taking log on both sides to get the final result:

$$f(y) = y^c \text{ ➔ } h(y) = c\,ln(y)$$

That concludes the Calculus-based discussion for this book, let us now discuss few important algebraic properties.

Firstly, a group G under operation $*$ is called a valid a group if and only if it satisfies the following properties:

(a) $\forall a, b \in G$, $a * b \in G$ also known as the closure property.
(b) $\forall a, b, c \in G$, $(a * b) * c = a * (b * c)$ also known as the commutative property.
(c) $\forall g \in G$, there must exist $g' \in G$ such that $g * g' = e$ also called as the inverse property.
(d) There must exist some $e \in G$, such that $\forall g \in G$, $g * e = g$ also known as the identity property.

Examples of valid groups are, $\langle \mathbb{Z}, + \rangle, \langle \mathbb{Q}, + \rangle, D_2$ and etc.

Let us now discuss few famous examples of groups.

The Klein-4 group can be defined as follows:

$$V = \{e, a, b, c\}$$

Here, e is the identity element and $\forall x \in V$, $|x| = 2$. The commutations between each element can be represented by the following table:

$*$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

It becomes quite easy to notices from the table that, $\forall x, y \in G$, $x * y = y * x$. Meaning, V is an abelian group.

Another takeaway being, $\forall x \in G$, $x * x = e$. Meaning, each element is its own self-inverse.

Completing a loop in any direction will result in identity element e, $abc = bca = acb = e$.

Moving on, we discuss the quaternion group given as follows:

$$Q = \{\pm 1, \pm i, \pm j, \pm k\}$$

The identity element of this group being 1. The commutations in this group are defined as follows:

(A) If the flow concurrency is persevered then the product is the next element in the flow. $i * j = k$, $j * k = i$ and $k * i = j$.
(B) If the flow concurrency is broken then the product is the negative of the next element in the flow. $j * i = -k$, $k * j = -i$ and $i * k = -j$.
(C) The product of an element with itself either results in 1 or -1. $i * i = j * j = k * k = -1$ and $-1 * -1 = 1 * 1 = 1$.
(D) Product of an element with its negative counterpart results in 1 or -1. $i * (-i) = j * (-j) = k * (-k) = 1$ and $1 * (-1) = -1$.
(E) Product of a direction vector element with a scalar result in itself or its negative counterpart. $i * (\pm 1) = \pm i$, $j * (\pm 1) = \pm j$, $k * (\pm 1) = \pm k$.

A quaternion in general is algebraically given as follows:

$$Q = a + bi + cj + dk$$

Such that $a, b, c, d \in \mathbb{R}$.

Moving ahead, we now discuss the dihedral groups. A list of dihedral groups is given down below:

(a) $D_2 = \{i, \rho, \mu\rho, \mu\}$
(b) $D_3 = \{i, \rho, \rho^2, \mu, \mu\rho, \mu\rho^2\}$
(c) $D_4 = \{i, \rho, \rho^2, \rho^3, \mu, \mu\rho, \mu\rho^2, \mu\rho^3\}$
(d) $D_5 = \{i, \rho, \rho^2, \rho^3, \rho^4, \mu, \mu\rho, \mu\rho^2, \mu\rho^3, \mu\rho^4\}$

Such that, i is the identity element. In general, Dihedral group of order n can be given as follows:

$$D_n = \left\{\mu^a\rho^b \middle| 0 \leq a \leq 1, 0 \leq b \leq n - 1\right\}$$

Few important computation-based properties for D_n are listed down below:

(a) $\mu^2 = i, \rho^n = i$
(b) $\rho^k\mu = \mu\rho^{n-k}$ (Assuming, $k < n$)
(c) $(\mu\rho)^2 = \mu\rho\mu\rho = \mu\mu\rho^{n-1}\rho = \mu^2\rho^n = i$
(d) $(\mu\rho^k)^n = \mu\rho^k\mu\rho^k = \mu\rho^k\rho^{n-k}\mu = i$ (Assuming, $k < n$)

Next up, we study about cyclic groups. A cyclic group of order n is defined as follows:

$$G = \left\{g^k \middle| 0 \leq k \leq n - 1\right\}$$

Such that, $|G| = n$ and 1 is the identity element of the group.

Inverse element $\exists g^x \in G$ such that $0 \leq x \leq n - 1$, is given as g^{n-x} as $g^x g^{n-x} = g^n = i$. Also, G is an abelian group and the proof is quite simplistic in nature.

Let $0 \leq a, b \leq n - 1$ and as a virtue $g^a, g^b \in G$. Thus:

$$g^a g^b = g^{a+b} = g^{b+a} = g^b g^a$$

This proves that G is an abelian group.

Also, if Cayley digraphs of G are constructed, $\forall |G| = n$ such that $n > 3$, the resulting graph will be a regular n-gon, where every edge represents g multiplied to the vertex element say g^x such that $0 \leq x \leq n - 1$. For example:

$$G' = \{1, g, g^2, g^3, g^4\}$$

Represents a regular pentagon. Similarly, $G_n = \left\{g^i \middle| \forall 0 \leq i \leq n-1\right\}$ represents a regular n-gon and it is essential to recall that n has to be strictly greater than or equal to 3.

Moving forward, group of integers of order n (such that $n > 1$) defined over addition, is a famous example of a cyclic group, defined as follows:

$$\mathbb{Z}_n = \{i | 0 \leq i \leq n-1\}$$

Such that, $|\mathbb{Z}_n| = n$ and $\mathbf{0}$ is the identity element. The addition scheme in the group is defined as:

$$a +_n b = (a + b) \, mod \, n$$

Such that, $\forall a, b \in \mathbb{Z}_n$. Proving that this group is abelian is also quite trivial $\forall a, b \in \mathbb{Z}_n$:

$$a +_n b = (a + b) \, mod \, n = (b + a) \, mod \, n = b +_n a$$

Speaking about inverses, $\exists x / \{0\} \in \mathbb{Z}_n$ the inverse of this element is given as $(n - x)$ because:

$$x +_n (n - x) = (x + n - x) \, mod \, n = n \, mod \, n = 0$$

For some final discussion, we will be quickly introducing ourselves to rings. A set R equipped with two binary operations say $+, \times$ hold the following conditions.

(A) $\langle R, + \rangle$ is an abelian group
(B) $\forall a, b, c \in R$, $(ab)c = a(bc)$ meaning multiplication must be associative.
(C) $\forall a, b, c \in R$, $(a + b)c = ac + ab$ meaning distribution property must hold

Examples of rings include, $\langle \mathbb{Z}, +, \times \rangle$, $\langle M_n(\mathbb{R}), +, \times \rangle$ and etc.

Most importantly, $\mathbb{Z}_n$ (such that $n > 1$) is a valid ring. The multiplication operation in $\mathbb{Z}_n$ is defined as follows $\forall a, b \in \mathbb{Z}_n$:

$$a *_n b = (ab) \, mod \, n$$

It becomes quite trivial to prove that ring is multiplicatively commutative $\forall a, b \in \mathbb{Z}_n$:

$$a *_n b = (ab)\, mod\, n = (ba)\, mod\, n = b *_n a$$

For instance, $P(x) = x^2 + (n-2)x + (n-3)$ will be reduced to the following in $\mathbb{Z}_n$:

$$P(x) = x^2 + nx - 2x + n - 3 = x^2 - 2x - 3$$

With this we conclude the appendix segment.

Important results

Here, we state some important Radon transform results, as per usual $X = [x_i]\ \forall i \leq n$ and also do note, in 2D cases $Rf(P,t)$ results are shown.

$f(X)$	$Rf(P,U)/Rf(P,t)$
e^{-x^2}	e^{-P^2}
$e^{-x^2-y^2}$	$\sqrt{\pi}e^{-P^2}$
e^{-X^TX}	$\left(\sqrt{\pi}\right)^{n-1} e^{-P^2}$
$H_n(x)H_m(y)e^{-x^2-y^2}$	$\sqrt{\pi}(cos\, t)^n(sin\, t)^m e^{-P^2} H_{n+m}(P)$
$e^{-\left(\frac{x}{a}\right)^2-\left(\frac{y}{b}\right)^2}$	$\frac{\sqrt{\pi}\lvert ab\rvert}{l} e^{-\left(\frac{P}{l}\right)^2}$
$\nabla f(X)$	$\frac{\partial^2}{\partial P^2} Rf(P,U)$
$f(x,y) = \begin{cases} A\,;\ x^2+y^2 \leq a \\ 0\,;\ x^2+y^2 > a \end{cases}$	$Rf(P,U) = \begin{cases} 2A\sqrt{a-P^2}; \lvert P\rvert \leq \sqrt{a} \\ 0\,; \lvert P\rvert > \sqrt{a} \end{cases}$
$\mu(r)$	$2\sqrt{1-P^2}\mu(P)$
$\delta(X)$	$\delta(P)$
$\delta(X-A)$	$\delta(P-P_0)$
$f(X)\delta(X-A)$	$f(A)\delta(P-P_0)$

Where, $l = \sqrt{(a\, cos\, t)^2 + (b\, sin\, t)^2)}$, $r = \sqrt{x^2+y^2}$, $P_0 = U^TA$ and $\mu(r)$ is the disk characteristic function.

References

[1] L. Debnath, Integral Transforms and Their Applications, Third edition, CRC Press, 2015

[2] A. Poularikas, The Transforms and Applications Handbook, Second edition, CRC Press, 2000

[3] A. Zentz, The Finite Radon Transform, B.S. Undergraduate Mathematics Exchange, Vol. 6, No. 1, Spring 2009.

[4] S. Helgason, The Radon Transform, Second Edition, Birkhäuser, 1999.

[5] F. Natterer, The Mathematics of Computerized Tomography, Society for Industrial and Applied Mathematics, 2001.

[6] E. T. Quinto, An Introduction to X-ray Tomography and Radon Transforms, Proceedings of Symposia in Applied Mathematics.

[7] G. Beylkin, The Inversion Problem and Applications of the Generalized Radon Transform.

[8] L. Hörmander, The Analysis of Linear Partial Differential Operators.

[9] G. B. Folland, Introduction to Partial Differential Equations.

[10] E. M. Stein and R. Shakarchi, Real Analysis: Measure Theory, Integration, and Hilbert Spaces, Princeton University Press.

Radon transforms, studied as a subject rather than as an isolated theory.
The book compiles together all foundational ideas, transform techniques,
Inversion methods and discretisation frameworks under one common roof.
The goal of this work is to showcase the beauty and rigor of Radon transforms.